Identifiability of a point defect model for oxide growth on AISI 347

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ARTICLE INFO

Keywords:
A. Stainless steel
B. Modelling studies
C. Oxidation
C. Passive films
C. High temperature corrosion

ABSTRACT

Mechanistic corrosion models are routinely fitted with four to six parameters to a few measurements and extrapolated across service lifetimes, yet whether the data determine them is rarely tested. We identify a reduced point defect model for the duplex oxide on Nb-stabilized AISI 347 in simulated boiling-water-reactor water from published depth profiles at three exposure times, by differentiable inversion graded against a closed-form reference. Profile likelihood, a 1000-member bootstrap and a prior-relaxation test agree that one of the five parameters is undetermined and a second only weakly so, and that 95% of the fit statistic rests on three chromium points. The kinetics track an empirical power law over the calibration window but diverge from it by a factor of 3.5 to 6.3 at ten years, outside the propagated uncertainty band. That separation, not the thickness it brackets, is what these data determine; an exposure near 3000 hours would resolve it.

1. Introduction

Nb-stabilized austenitic stainless steel AISI 347 (X6CrNiNb18-10) is used in boiling-water-reactor (BWR) internals and piping for its resistance to weld-decay sensitization, yet the evolution of its passive oxide film under reactor-representative hydrothermal water has so far had no mechanistic predictive model. Current practice for predicting oxide behavior on light-water-reactor internals relies either on operational water-chemistry criteria framed around threshold electrochemical corrosion potential (ECP) and dissolved-oxygen levels (Macdonald, 1992b; Lin, Kim, Niedrach and Ramp, 1996; Yeh, Macdonald and Motta, 1995; Yeh and Macdonald, 1996), which say nothing about oxide thickness or composition, or on data-driven machine-learning models of crack-growth rate (Falaakh and Bahn, 2025), which are equally silent on the underlying oxide chemistry.

The duplex oxide that austenitic stainless steels develop in high-temperature water (a Cr-rich inner spinel layer growing by solid-state transport beneath an Fe-rich outer layer formed by precipitation) is well established for the 304/316 grades (Robertson, 1991; Stellwag, 1998; Ziemniak and Hanson, 2002; Kuang, Wu and Han, 2010; Terachi, Yamada, Miyamoto, Arioka and Fukuya, 2008; Kim, 1999; Lister, Davidson and McAlpine, 1987), and the point defect model (PDM) of Chao, Lin, and Macdonald (1981; 1981; 1992a; 2011), in the tradition of high-field oxidation theory (Cabrera and Mott, 1949), provides the standard mechanistic description: vacancy-mediated transport under a high interfacial field, yielding closed-form thickness-versus-time laws tied to interfacial reaction kinetics. Li, Macdonald, Yang, Qiu and Wang (2020) extended this framework to supercritical water. For AISI 347 specifically, Veile, Hirpara, Lackmann and Weihe (2024) recently provided the first detailed characterization of the duplex oxide developed in simulated BWR hydrothermal water, resolving Cr-rich barrier and Fe-rich outer layers by energy-dispersive X-ray (EDX) depth profiling at three exposure times; their analysis stops at empirical per-element power-law fits, with no shared physical parameter set connecting the layers and no assessment of how far those fits can be extrapolated.

This paper adapts the supercritical-water PDM formulation (Li et al., 2020) to the subcritical hydrothermal-water regime and identifies its kinetics from the Veile et al. dataset with a differentiable, physics-informed inversion framework. The alloy specificity of the result lies in the data, not the model structure: the reaction network contains nothing Nb-specific, and what is delivered is the first PDM identification against the only depth-profile dataset available for this alloy and environment. The identification is the vehicle rather than the destination. The question this paper is built around is a methodological gap that recurs across four decades of PDM parameter estimation. Whether calibrated against electrochemical impedance spectra (Bojinov, Kinnunen, Lundgren and Wikmark, 2005; Zhao, Chen, Qiu,

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Sun and Macdonald, 2024; Fattah-alhosseini, 2016) or against depth-profile data, four-to-six-parameter models are routinely fit to three-to-six data points, and while parameter confidence intervals and correlations are sometimes reported (Berthier, Diard, Pronzato and Walter, 1996), we are not aware of a formal profile-likelihood or bootstrap identifiability characterization of a PDM fit. The distinction between "the model fits" and "the data constrain the parameters" becomes consequential exactly when the fitted model is meant to extrapolate beyond its calibration window, as here. We close this gap with tools mature in adjacent fields (Raue, Kreutz, Maiwald, Bachmann, Schilling, Klingmüller and Timmer, 2009; Efron, 1979; Wieland, Hauber, Rosenblatt, Tönsing and Timmer, 2021), standard in systems biology (Gábor, Villaverde and Banga, 2017) and increasingly applied to battery models (Laue, Röder and Krewer, 2021), built directly into the inversion rather than bolted on afterward.

What that analysis returns, and what organizes the rest of this paper, is a single answer arrived at from four directions. Of the five parameters the model needs, three exposure times determine three, though the third is sharp in profile and not under resampling; one is set by its prior and one only to within an order of magnitude. Ninety-five percent of the fit statistic rests on the three chromium points and eighty percent on one of them, and the residual pattern is systematic in precisely the curvature that governs extrapolation. The spatial extension and the two candidate mobility models each fail an identifiability test for the same reason, and the nickel-zone closure fails on shape for it too. The consequence is that the quantity these data determine is not the ten-year thickness but the separation between the mechanistic and empirical extrapolations (a factor of 3.5 to 6.3 at ten years), and that no reanalysis of three exposures can sharpen it, while one designed exposure of roughly 3000 hours would. Too few independent exposure times is the paper's recurring finding, and each section below reaches it independently.

Two results fall outside that thread and are stated here so they are not mistaken for asides. Extended spatially with no additional free parameters, the identified kinetics predict the measured composition profile, and a structural argument shows why composition data cannot constrain the defect diffusivity however many of them are taken. And solving Poisson's equation together with the defect transport, rather than imposing the field, shows that the constant-field closure on which the PDM's field strength rests is not self-consistent at the defect concentrations the model implies.

Those results are produced by a differentiable inversion framework built on a physics-informed neural solver (Raissi, Perdikaris and Karniadakis, 2019; Karniadakis, Kevrekidis, Lu, Perdikaris, Wang and Yang, 2021), the neural spectral-element method (NSEM) (Tetsassi Feugmo and Pankaczy, 2026). Kinetic parameters are recovered by a hard-constrained inversion in which the trajectories are generated by differentiable Runge–Kutta integration, so the governing equations are satisfied exactly by construction; a classical least-squares solver is an independent reference, and the two agree to within 0.6% on every parameter and to identical χ^2. The same machinery emulates the forward model with two neural backbones and, in a soft-constrained joint field-and-parameter mode, extends to the spatial transport problem.

It is worth being precise about what that machinery contributes. The property doing the statistical work is *differentiability* rather than the neural representation: exact parameter gradients through a vectorized solver are what make profile likelihood, a thousand-member bootstrap, prior-relaxation scans and a dense operating-envelope grid affordable, and closed-form models with automatic differentiation share that property. The neural representation becomes necessary only where no closed form or cheap forward solve exists, which is not the case here: classical methods reproduce every result in this paper, and in the spatial case more accurately. That is the point of the test case rather than a defect of it: a framework can be graded only where an exact reference exists, and grading it is the prerequisite for trusting it where none does, a bound made explicit in Section 4. Our prior work demonstrated this machinery on the PDM against synthetic benchmarked data, cataloging four training-time failure modes (Farooqi, Bösing and Tetsassi Feugmo, 2026); the present paper is its first application to real experimental data, and in carrying it out we document and remedy a fifth, specific to sparse real data: joint field-and-parameter training can settle into an equilibrium that satisfies the data loss while leaving the governing equation only approximately satisfied, exposed only by re-solving the recovered parameters through an exact forward solve. Loss-balancing pathologies of this general kind are a recognized theme in physics-informed training (Wang, Yu and Perdikaris, 2022; Krishnapriyan, Gholami, Zhe, Kirby and Mahoney, 2021), and a contemporaneous study reports a closely related diagnosis in an unrelated domain (Almanstötter, Vetter and Iber, 2025); the contribution here is its documentation, its remedy, and a self-consistency check that travels with the framework.

The paper is organized methods-first: Section 2 derives the reduced model step by step and sets out the data, the statistical machinery and the solver; Section 3 carries the argument above; and the Discussion reads the identified parameters mechanistically before bounding how far that reading can be taken.

2. Model and methods

2.1. Reduced-model derivation

The model identified in this paper is a reduced form of the point defect model in the formulation Li et al. (2020) developed for supercritical water, adapted here to subcritical hydrothermal water. Every parameter reported later is an output of the reduction rather than an elementary rate constant, so the derivation is set out step by step below and each symbol is defined where it first appears. The full reaction tables, the activated-complex derivation of the rate-constant form, and known errata in the source formulations (Li et al., 2020) are given in the Supplementary Information (SI, Section S1).

Step 1: which reactions can move an interface. The oxide Veile et al. (Veile et al., 2024) characterize on X6CrNiNb18-10 is a duplex scale (Fig. 1a): a chromium-rich, nanocrystalline spinel barrier layer of thickness L_{bl} growing into the metal, and discrete iron-rich magnetite crystals forming a porous outer layer of thickness L_{ol}, with a nickel enrichment zone at the receding metal/barrier-layer interface. The kinetics derived here treat that nickel zone as an inert consistency check; a dedicated transport model for it is developed in Section 3.5. Reactions occur at three interfaces: metal/barrier layer (m/bl), barrier layer/outer layer (bl/ol), and outer layer/water. Of the fourteen reactions in the network (eleven numbered, three of them with primed variants), only four displace a boundary. Reaction 3, $m \rightarrow M_M + (\chi/2)\,V_O^{\bullet\bullet} + \chi e'$, in which a metal atom m enters the oxide as a lattice cation M_M while generating oxygen vacancies $V_O^{\bullet\bullet}$ (charge number χ, defined in Step 3) and electrons e', grows the barrier layer into the metal; Reactions 10 and $10'$ convert or dissolve barrier-layer oxide; Reaction 11 dissolves or further oxidizes outer-layer oxide. The other ten are lattice-conservative: they carry cations across the barrier layer and deposit them in the outer layer without themselves moving a boundary. The thickness equations therefore contain only these four rates.

Step 2: the potential distribution and the constant-field postulate. Let V be the potential difference between metal and solution and $\phi_{\mathrm{m/bl}}$, $\phi_{\mathrm{bl/e}}$ the drops at the metal/barrier-layer and barrier-layer/water interfaces. The PDM postulates that the electric field inside the barrier layer is constant at ε_f, independent of both V and L_{bl}, and that the outer drop is linear in potential and pH,

$$\phi_{\mathrm{bl/e}} = \alpha V + \beta\,\mathrm{pH} + \phi^0, \tag{1}$$

where $\alpha = \partial\phi_{\mathrm{bl/e}}/\partial V$ is the polarizability of that interface, $\beta = \partial\phi_{\mathrm{bl/e}}/\partial\,\mathrm{pH} < 0$, and ϕ^0 is the drop at $V = \mathrm{pH} = L_{\mathrm{bl}} = 0$. Summing the drops across the stack, $V = \phi_{\mathrm{m/bl}} + \varepsilon_f L_{\mathrm{bl}} + \phi_{\mathrm{bl/e}}$, so that

$$\phi_{\mathrm{m/bl}} = (1-\alpha)V - \varepsilon_f L_{\mathrm{bl}} - \beta\,\mathrm{pH} - \phi^0. \tag{2}$$

The term that carries the kinetics is $-\varepsilon_f L_{\mathrm{bl}}$: the potential available to drive the metal/barrier-layer reactions falls linearly as the film thickens.

Step 3: the rate constants inherit that thickness dependence. Activated-complex theory with partial charge transfer gives every interfacial rate constant the form

$$k_i = k_i^0\, e^{a_i V}\, e^{b_i L_{\mathrm{bl}}}\, e^{c_i\,\mathrm{pH}}, \tag{3}$$

with k_i^0 the standard rate constant of Reaction i and a_i, b_i, c_i fixed by which interfacial drop that reaction sees (SI, Table S5). Reactions at bl/ol and at the outer surface see $\phi_{\mathrm{bl/e}}$, which by Eq. (1) carries no L_{bl} dependence, so $b_i = 0$ for all of them. The m/bl reactions see $\phi_{\mathrm{m/bl}}$ and therefore inherit Eq. (2); for the growth reaction this gives

$$b_3 = -\alpha_3 \chi\gamma\varepsilon_f < 0, \tag{4}$$

in which α_3 is the transfer coefficient of Reaction 3 (dimensionless), $\chi = 8/3$ the spinel-averaged cation charge number, and $\gamma = F/RT = 22.6\ \mathrm{V}^{-1}$ at 240°C, with F the Faraday constant, R the gas constant and T the absolute temperature. The negative sign is the kinetic content of the model: growth decelerates exponentially in the film's own thickness, which is what produces direct-logarithmic kinetics and, whenever a destruction term is nonzero, a finite steady-state thickness.

Step 4: the barrier-layer thickness equation. Only the non-conservative reactions of Step 1 move the barrier layer's boundaries, so its thickness obeys

$$\frac{\mathrm{d}L_{\mathrm{bl}}}{\mathrm{d}t} = \Omega_{\mathrm{bl}}\left(k_3 - k_d\right)\frac{\rho_{\mathrm{bl}}^0}{\rho_{\mathrm{bl}}}, \qquad k_d := k_{10} C_{\mathrm{O}}^{q} + k_{10'}\left(C_{\mathrm{H}^+}/C_{\mathrm{H}^+}^0\right)^s, \tag{5}$$

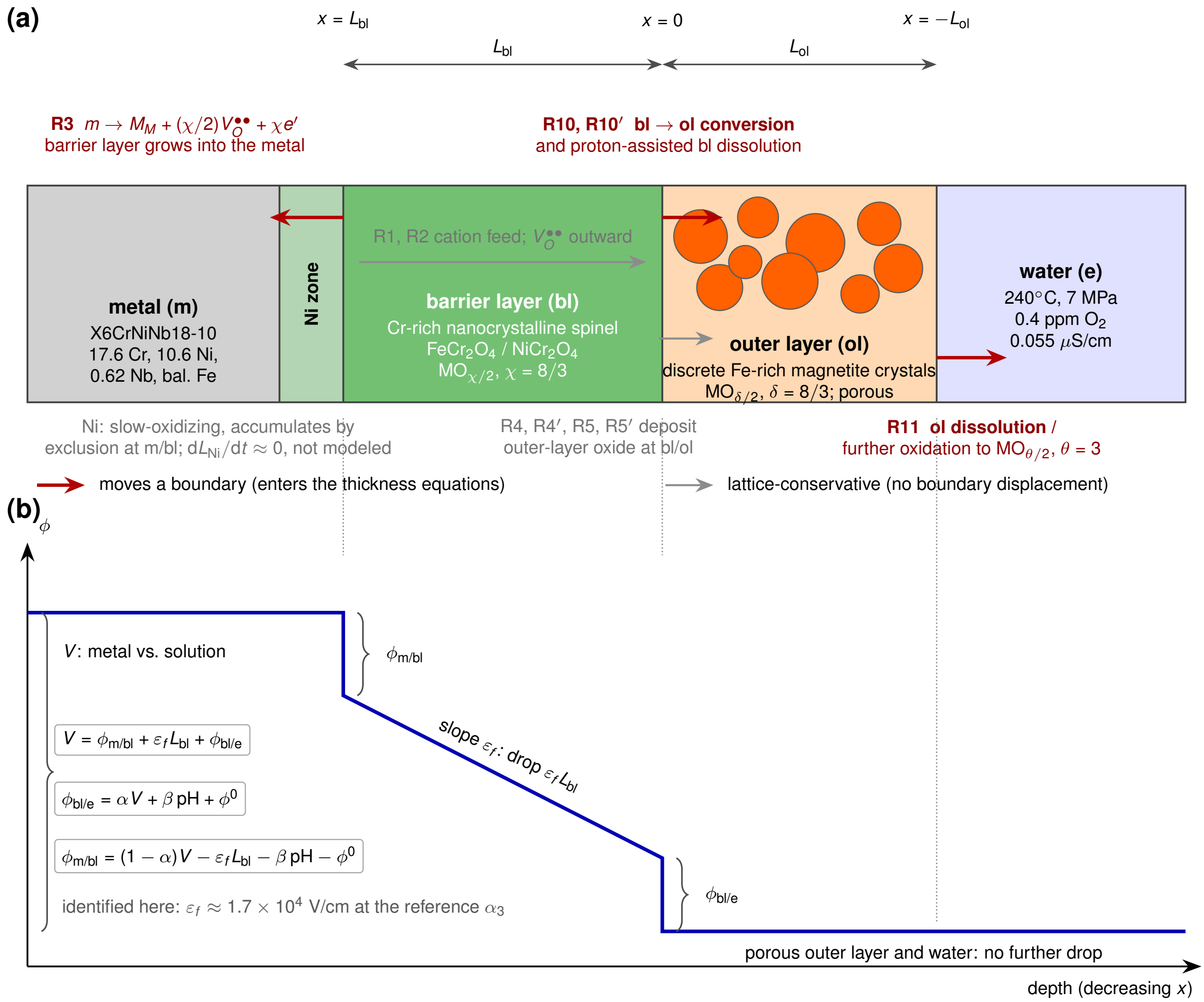

Figure 1: Duplex-oxide model schematic, not to scale. (a) Layer structure, drawn left to right in order of increasing distance from the metal: metal | Ni enrichment zone | Cr-rich nanocrystalline spinel barrier layer | porous Fe-rich outer layer of discrete magnetite crystals | hydrothermal water. Interface positions carry the coordinate convention of the SI (Section S1.1), and the two state variables L_{bl} and L_{ol} are dimensioned above the stack. Each reaction is drawn at the interface where it acts, in red if it displaces a boundary and therefore enters Eqs. (10)–(11), in gray if it is lattice-conservative: Reaction 3 grows the barrier layer into the metal, Reactions 10 and 10′ convert and dissolve it at the barrier-layer/outer-layer interface, and Reaction 11 destroys the outer layer at the outer surface. (b) Potential distribution across the same stack under the constant-field postulate. The abscissa is aligned with (a) and the ordinate is the electrostatic potential, so the decomposition $V = \phi_{\mathrm{m/bl}} + \varepsilon_f L_{\mathrm{bl}} + \phi_{\mathrm{bl/e}}$ reads off the profile directly: a step at each of the two interfaces that carries a drop, and between them the constant slope ε_f of Eq. (2). The porous outer layer carries no further drop.

where Ω_{bl} is the molar volume of the barrier oxide per cation and $\rho^0_{\mathrm{bl}}/\rho_{\mathrm{bl}}$ a porosity correction (set to unity for both layers throughout, for the reason given in Step 6), C_{O} the dissolved-oxygen concentration, $C_{\mathrm{H^+}}/C^0_{\mathrm{H^+}}$ the proton activity ratio, and q and s the corresponding reaction orders. The lumped rate k_d is the total barrier-layer destruction rate. Substituting Eq. (3) for k_3 at fixed V and pH puts Eq. (5) in the two-parameter working form used throughout, with

$$A_{\mathrm{bl}} = \Omega_{\mathrm{bl}}\, k_3^0\, e^{a_3 V}\, e^{c_3\,\mathrm{pH}}\, \frac{\rho^0_{\mathrm{bl}}}{\rho_{\mathrm{bl}}}, \qquad C_{\mathrm{bl}} = \Omega_{\mathrm{bl}}\, k_d\, \frac{\rho^0_{\mathrm{bl}}}{\rho_{\mathrm{bl}}}. \tag{6}$$

Step 5: the outer layer before coupling. The outer layer grows from cations transmitted through the barrier layer, from barrier oxide destroyed by Reactions 10/10′ and re-precipitated as outer oxide, and shrinks by dissolution:

$$\frac{\mathrm{d}L_{\mathrm{ol}}}{\mathrm{d}t} = \Omega_{\mathrm{ol}} \left(k_1 C_{V_M}^L + k_2 + k_d - k_{11} C_{\mathrm{O}}^r \right) \frac{\rho_{\mathrm{ol}}^0}{\rho_{\mathrm{ol}}}, \tag{7}$$

in which Ω_{ol} and $\rho_{\mathrm{ol}}^0/\rho_{\mathrm{ol}}$ are the outer layer's molar volume per cation and porosity correction, $k_1 C_{V_M}^L$ and k_2 are the cation fluxes arriving by the vacancy and interstitial paths respectively ($C_{V_M}^L$ being the cation-vacancy concentration at the m/bl interface), and $k_{11} C_{\mathrm{O}}^r$ is the outer-layer dissolution rate with order r. Following Ref. (Li et al., 2020), the whole of k_d is credited to the outer layer, which slightly overstates its supply because the proton-assisted route $k_{10'}$ ejects cations into solution rather than into oxide; in this ultrapure-water environment k_d is fitted to zero in any case. Written this way the two layers are still independent, and the six rates in Eqs. (5) and (7) are far more than three exposure times can determine.

Step 6: the constant-volume constraint collapses the coupling. Marker experiments place the bl/ol interface at the original metal surface, so the oxide grows into the metal rather than outward from it: metal is consumed at exactly the rate the barrier layer advances, $\mathrm{d}L_m/\mathrm{d}t = \mathrm{d}L_{\mathrm{bl}}/\mathrm{d}t$, where L_m is the thickness of metal consumed, $\mathrm{d}L_m/\mathrm{d}t = \Omega_m(k_1 C_{V_M}^L + k_2 + k_3)$ and Ω_m is the metal's molar volume per atom. Equating the two and using Eq. (5) gives the constant-volume condition

$$k_1 C_{V_M}^L + k_2 + k_3 = \mathrm{PBR}\,(k_3 - k_d), \qquad \mathrm{PBR} := \Omega_{\mathrm{bl}}/\Omega_m, \tag{8}$$

PBR being the Pilling–Bedworth ratio of the barrier oxide. The metal is dense, so the left-hand side of Eq. (8) carries no porosity correction while the right-hand side inherits $\rho_{\mathrm{bl}}^0/\rho_{\mathrm{bl}}$ from Eq. (5). Writing Eq. (8) without that factor is therefore the dense-oxide convention $\rho_{\mathrm{bl}}^0/\rho_{\mathrm{bl}} = \rho_{\mathrm{ol}}^0/\rho_{\mathrm{ol}} = 1$, and equal porosity between the two layers is not sufficient for it: carrying both corrections through the substitution below leaves the outer-to-barrier coefficient as $(\Omega_{\mathrm{ol}}/\Omega_{\mathrm{bl}})(\rho_{\mathrm{ol}}^0/\rho_{\mathrm{ol}})(\mathrm{PBR} - \rho_{\mathrm{bl}}/\rho_{\mathrm{bl}}^0)$, which collapses to the form used below only when both are unity. The convention bears on the physical quantities recovered in Step 9, not on the fitted parameters, which absorb it. Substituting Eq. (8) into Eq. (7), the two destruction terms combine exactly, $(\mathrm{PBR} - 1)k_3 - \mathrm{PBR}\,k_d + k_d = (\mathrm{PBR} - 1)(k_3 - k_d)$, leaving the cation fluxes eliminated and the outer-layer growth rate an exact affine function of the barrier-layer growth rate,

$$\frac{\mathrm{d}L_{\mathrm{ol}}}{\mathrm{d}t} = (\mathrm{PBR} - 1)\frac{\Omega_{\mathrm{ol}}}{\Omega_{\mathrm{bl}}}\frac{\mathrm{d}L_{\mathrm{bl}}}{\mathrm{d}t} - \Omega_{\mathrm{ol}}\, k_{11} C_{\mathrm{O}}^r \frac{\rho_{\mathrm{ol}}^0}{\rho_{\mathrm{ol}}}. \tag{9}$$

This is the step that ties the two layers to one growth law, and it is what makes the mechanistic and empirical descriptions extrapolate differently in Section 3.6.

Step 7: reduction to apparent parameters. At open circuit V and pH are constant over an exposure (the water is neutral, unbuffered and of very low conductivity; the neutral pH at 240°C is approximately 5.6), so the factor $e^{a_3 V + c_3\,\mathrm{pH}}$ in Eq. (6) is a constant multiplier that thickness-versus-time data cannot separate from k_3^0. The same is true of the molar volumes and porosity corrections. What the data can determine is therefore not the elementary rate constants but the lumped groups of Eqs. (6), (8) and (9), giving the apparent-parameter system fitted in this paper:

$$\frac{\mathrm{d}L_{\mathrm{bl}}}{\mathrm{d}t} = A_{\mathrm{bl}}\, e^{b_3 L_{\mathrm{bl}}} - C_{\mathrm{bl}}, \qquad L_{\mathrm{bl}}(0) = L_0, \tag{10}$$

$$\frac{\mathrm{d}L_{\mathrm{ol}}}{\mathrm{d}t} = \mathrm{PBR}_{\mathrm{eff}}\,\frac{\mathrm{d}L_{\mathrm{bl}}}{\mathrm{d}t} + C_x, \qquad L_{\mathrm{ol}}(0) = L_{\mathrm{ol},0}, \tag{11}$$

where t is exposure time (h) and $L_{\mathrm{bl}}(t)$, $L_{\mathrm{ol}}(t)$ are the two layer thicknesses (nm). The five rate parameters are $A_{\mathrm{bl}} > 0$ (nm/h), the apparent growth prefactor of Eq. (6), which absorbs the potential- and pH-dependent part of the growth rate constant; $b_3 < 0$ (nm^{-1}), the apparent field-related exponent of Eq. (4), which sets how steeply growth decelerates as the film thickens; $C_{\mathrm{bl}} \geq 0$ (nm/h), the barrier-layer destruction rate;

$$\begin{aligned} \mathrm{PBR}_{\mathrm{eff}} &:= (\mathrm{PBR} - 1)\,\frac{\Omega_{\mathrm{ol}}}{\Omega_{\mathrm{bl}}} \qquad && \text{(dimensionless)}, \\ C_x &:= -\Omega_{\mathrm{ol}}\, k_{11} C_{\mathrm{O}}^r \frac{\rho_{\mathrm{ol}}^0}{\rho_{\mathrm{ol}}} \;\leq\; 0 \quad && \text{(nm/h)}, \end{aligned} \tag{12}$$

the apparent outer-to-barrier growth-rate ratio and the outer-layer dissolution term, the latter written with a sign convention such that a negative value removes outer-layer material. The initial conditions $L_{\mathrm{bl}}(0) = L_0$ and $L_{\mathrm{ol}}(0) = L_{\mathrm{ol},0}$ (both nm) complete the seven-parameter set.

Step 8: closed-form solutions. Both equations integrate analytically at constant parameters. The barrier layer gives

$$L_{\mathrm{bl}}(t) = L_0 - \frac{1}{b_3} \ln\left[1 + \frac{A_{\mathrm{bl}}}{C_{\mathrm{bl}}}\, e^{b_3 L_0} \left(e^{-b_3 C_{\mathrm{bl}} t} - 1\right)\right] - C_{\mathrm{bl}}\, t, \tag{13}$$

which for $C_{\mathrm{bl}} \to 0$ reduces to the direct-logarithmic law $L_{\mathrm{bl}}(t) = L_0 - b_3^{-1} \ln(1 - b_3 A_{\mathrm{bl}} e^{b_3 L_0} t)$ and for $C_{\mathrm{bl}} > 0$ saturates at a finite $L_{\mathrm{bl,ss}}$; those two branches are the M4 and M5 variants compared in Section 3.6. Because Eq. (11) is affine in $\mathrm{d}L_{\mathrm{bl}}/\mathrm{d}t$, the outer layer follows by direct integration with no further approximation,

$$L_{\mathrm{ol}}(t) = L_{\mathrm{ol},0} + \mathrm{PBR}_{\mathrm{eff}} \left(L_{\mathrm{bl}}(t) - L_0\right) + C_x\, t. \tag{14}$$

Equations (13)–(14) are evaluated in log space so that they remain numerically exact in both the $|b_3|L \ll 1$ and $|b_3|L = O(1)$ limits (SI, Section S1.7); this closed form, not a numerical integration, is what the fits and the bootstrap evaluate, which is why a thousand refits are affordable.

Step 9: recovering physical quantities. The fitted apparent parameters map back to the underlying physics through

$$\varepsilon_f = \frac{-b_3}{\alpha_3 \chi \gamma}, \qquad \mathrm{PBR} = 1 + \mathrm{PBR}_{\mathrm{eff}} \frac{\Omega_{\mathrm{bl}}}{\Omega_{\mathrm{ol}}}, \qquad k_d = \frac{C_{\mathrm{bl}}}{\Omega_{\mathrm{bl}}}, \tag{15}$$

each inverting one of Eqs. (4), (12) and (6). The second and third both inherit the dense-oxide convention of Step 6, the second because Eq. (9) is derived under it and the third because the porosity correction does not cancel out of Eq. (6); the first involves no porosity correction and needs neither. The first is evaluated with b_3 converted to the cgs units in which field strengths are conventionally quoted ($1\ \mathrm{nm}^{-1} = 10^7\ \mathrm{cm}^{-1}$), so that $b_3 = -0.0125\ \mathrm{nm}^{-1} = -1.25\times10^5\ \mathrm{cm}^{-1}$ gives ε_f in $\mathrm{V\,cm}^{-1}$. Only the first requires an assumption the thickness data cannot supply, the transfer coefficient α_3, which is why the field strength is reported over a range of α_3 in Section 4 and why the identification is stated in terms of b_3 rather than ε_f.

Two independent checks establish that the closed-form solutions are exact solutions of the reduced system; the exactness of the reduction itself rests on the constant-volume closure of Step 6 and the equal-porosity convention. The closed-form solutions (13)–(14) agree with Radau numerical integration of Eqs. (10)–(11) to better than 6×10^{-11} nm (SI, Fig. S1), and the affine outer-layer form (14) agrees with the algebraically heavier closed form of Ref. (Li et al., 2020) to better than 1×10^{-10} nm.

2.2. Experimental data

The reduction leaves seven apparent parameters, and everything that can be asked of them is set by the one dataset that exists for this alloy in this environment. Veile et al. (2024) exposed X6CrNiNb18-10 coupons to simulated BWR hydrothermal water (240°C, 7 MPa, 0.4 ppm dissolved O_2) for 72, 168, and 480 h and measured element-resolved oxide layer thicknesses by focused-ion-beam cross-sections with EDX line scans. Table 1 reproduces the digitized layer-thickness means used as the inversion target throughout this paper. The digitization was checked against their own reported empirical fits: a scan-level linear-space least-squares refit of $L = k\,t^n$ to our digitized chromium data reproduces their published coefficients to four significant figures ($k = 6.522$ versus their 6.521, $n = 0.4964$ versus 0.4964), which bounds the digitization error well below the scan-to-scan scatter. For the iron layer the reported scan-to-scan physical scatter of the discrete-crystal outer layer exceeds digitization precision, so their reported layer means are used as the authoritative data. No synthetic or simulated data enters the kinetic identification; synthetic data appears only as an explicitly labeled recovery test (Section 3.3).

2.3. Statistical inversion and identifiability analysis

Six layer means at three exposure times against a five-parameter model is the ratio that dictates both the objective below and the identifiability machinery carried alongside it: at this sample size a good fit is not evidence that the

Table 1
Layer-thickness data of Veile et al. (2024) (their Fig. 9) used in this work: mean ± population standard deviation over n EDX line scans. The fits use standard errors $\sigma = \mathrm{SD}/\sqrt{n}$. The Ni enrichment zone is not a growing PDM layer and enters only the analysis of Section 3.5.

Layer	t (h)	mean (nm)	SD (nm)	n
Cr (barrier)	72	37.3	7.7	3
	168	98.3	21.7	4
	480	134.3	5.4	3
Fe (outer)	72	143.3	111.5	3
	168	225.5	142.7	4
	480	240.3	172.5	3
Ni (zone)	72	31.3	3.4	3
	168	43.0	11.1	4
	480	36.0	5.0	3

parameters are determined, and the two questions have to be asked separately. All fits minimize the weighted least-squares objective

$$\Phi(\theta) = \sum_{i=1}^{3} \frac{\left[L_{\mathrm{bl}}(t_i;\theta) - \bar{L}_{\mathrm{Cr},i}\right]^2}{\sigma_{\mathrm{Cr},i}^2} + \sum_{i=1}^{3} \frac{\left[L_{\mathrm{ol}}(t_i;\theta) - \bar{L}_{\mathrm{Fe},i}\right]^2}{\sigma_{\mathrm{Fe},i}^2} + \left(\frac{\ln L_0 - \ln 2\,\mathrm{nm}}{0.60}\right)^2, \tag{16}$$

where θ is the parameter vector being estimated, the index i runs over the three exposure times $t_i \in \{72, 168, 480\}$ h, $L_{\mathrm{bl}}(t_i;\theta)$ and $L_{\mathrm{ol}}(t_i;\theta)$ are the closed-form model thicknesses at those times, $\bar{L}_{\mathrm{Cr},i}$ and $\bar{L}_{\mathrm{Fe},i}$ are the measured layer means and $\sigma_{\mathrm{Cr},i}$, $\sigma_{\mathrm{Fe},i}$ their standard errors (Table 1), and the final term is a Gaussian prior in log-parameter space on the initial barrier-layer thickness L_0, centered on the 1–5 nm air-formed film on high-Cr stainless steel with a log-space width $\sigma_{\ln L_0} = 0.60$. It is the only prior in the objective, and it is carried because the exposure-time data give L_0 no interior optimum at all: without it the estimate runs to zero (Section 3.2). In particular no prior is placed on $\mathrm{PBR}_{\mathrm{eff}}$, which is determined by the likelihood alone throughout; the stoichiometric reasoning that decides what that value can be compared against is taken up in Section 4. The classical minimization uses Levenberg–Marquardt least squares in log-parameter space (which enforces positivity by construction), warm-started and verified against grid restarts; convergence tolerances are 10^{-8}. The equivalent hard-mode physics-informed inversion (Section 2.4) minimizes the same objective with the trajectories generated by differentiable Runge–Kutta integration. The two are independent routes to the same optimum rather than a primary and a check, and Section 3.3 reports how closely they agree; the deterministic parameter values and curves given throughout (Figs. 2–5) are the converged optimum common to both.

Profile-likelihood scans fix one parameter on a grid of ±1.5 natural-log units about the optimum (61 points) and re-optimize all others under the full objective (16); the $\Delta\chi^2$ values reported are computed over the data residuals alone, so a prior-dominated parameter is exposed as flat rather than masked. The parametric bootstrap resamples each layer mean from a Gaussian with its standard error (1000 members, all refits converged; positivity is guaranteed by the log-space parametrization) and refits the full objective.

Because one parameter is prior-set and a second only weakly determined, the number of data-determined parameters is between three and five, and the residual degrees of freedom for the accepted five-parameter model are correspondingly between 1 and 3 for the six data points. Goodness-of-fit statements are therefore given for both endpoints of this bracket rather than a single nominal degree-of-freedom count.

2.4. Physics-informed spectral-element solver

The engine that minimizes that objective, and the forward solver used for the spatial extension later, are the same piece of machinery, described here once. The physics-informed forward and inverse solves use the neural spectral-element method (NSEM) (Tetsassi Feugmo and Pankaczy, 2026), in which each field is evaluated only at fixed Legendre–Gauss–Lobatto quadrature nodes and derivatives are taken with precomputed spectral differentiation matrices, giving a deterministic loss that L-BFGS can drive to residuals well below the accuracy floor of random-collocation training. The solver was applied to the PDM family in our prior work (Farooqi et al., 2026). Each field is represented by an element network evaluated on a collocation node set, physical residuals are enforced at the nodes through a first-integral or collocation formulation, loss terms are combined by a balanced-residual dynamic reweighting aggregator, and optimization proceeds in two phases (Adam, then L-BFGS). Full architecture, loss definitions, and hyperparameters for every configuration are tabulated in the SI (Section S4), which also summarizes the workflow in one diagram (Fig. S4).

Three configurations produce the results reported here; a fourth, a pure forward solve, is used only for the backbone verification described below. The three are: a "hard" zero-dimensional inverse, in which the kinetic parameters are the only learnable quantities and the trajectories are generated by differentiable Runge–Kutta integration, so Eqs. (10)–(11) are satisfied exactly by construction; a "soft" zero-dimensional inverse, in which field networks represent the trajectories directly and are trained jointly with the parameters against a combined data-and-physics loss (the configuration in which the failure mode of Section 3.3 is exposed); and the spatial extension of Section 3.4, where no differentiable-integrator shortcut exists and the soft-mode field representation is the only option, verified against an independent Newton solve on the same node set.

Every configuration is checked against an independent reference before its results are used. The classical baseline itself is established to machine precision (Section 2.1). Two neural backbones, a multilayer perceptron (MLP) and a Kolmogorov–Arnold network (KAN) (Liu, Wang, Vaidya, Ruehle, Halverson, Soljačić, Hou and Tegmark, 2024), are evaluated in the same harness to test backbone sensitivity, both to the same acceptance criterion (relative L^∞ error $\leq 0.5\%$ against the closed form). All results are backed by an automated test suite (25 tests) checking parity between closed-form, Newton, and neural solutions; the suite ships with the code release (see Code availability).

3. Results

The results follow the order the argument requires. We first establish that the reduced model of Section 2.1 can be identified from these data at all, and which of its variants the data select (Section 3.1). We then ask the question that decides what the identification is worth, namely which parameters the three exposure times actually determine (Section 3.2), and validate the inversion machinery itself against the exact reference this problem happens to admit, which is also where the failure mode we document is exposed (Section 3.3). The remaining sections spend the identified kinetics: on a spatial extension that predicts composition with no further free parameters (Section 3.4), on the one feature of the scale those kinetics fail to describe (Section 3.5), and on the long-time and operating-envelope predictions that are the reason for fitting a mechanistic model in the first place (Section 3.6).

3.1. Differentiable inversion and the model ladder

We identify the reduced kinetics by minimizing a weighted least-squares objective (Section 2.3, Eq. (16)), solving the inverse problem two independent ways: a hard-constrained physics-informed neural inversion, in which the trajectories are produced by differentiable Runge–Kutta integration so that Eqs. (10)–(11) hold exactly, and a classical Levenberg–Marquardt solver. Section 3.3 grades the two against each other; they agree closely enough that a single identified parameter set is reported, and the ladder below is used only to decide model structure.

We fit the reduced system (10)–(11) to the joint Cr and Fe thickness means (Table 1) in a nested model ladder (Table 2), each variant adding degrees of freedom to test whether the flexibility is supported by the data. The four-parameter core model (M1, $C_{\mathrm{bl}} = C_x = 0$, $L_{\mathrm{ol},0} = 0$) reaches $\chi^2 = 7.15$ only by driving $\mathrm{PBR}_{\mathrm{eff}}$ to 2.43, and still cannot reproduce the disproportionately thick early iron layer under strict constant-volume coupling from zero initial thickness; adding a free dissolution rate (M2) does not help, because the fit drives that rate to zero. Releasing the coupling entirely in the six-parameter M3 variant does not produce a better model but a degenerate one, and with no prior to hold it back the degeneracy is now the interior optimum rather than something a profile has to expose: the growth prefactor and dissolution rate diverge together to $\sim 10^3$ nm/h while the field parameter collapses by four orders of magnitude to -4.4×10^{-6} nm^{-1}, buying the lowest data χ^2 in the ladder (4.11) through an unphysical linear-growth escape route. Its ten-year barrier-layer prediction is 155 nm against M4's 535 nm, so the χ^2 ordering and the physical ordering disagree outright. That M3 wins on fit and loses on physics is the clearest statement in this paper of why χ^2 cannot by itself select a model on six data points. Parameter correlations of this kind have long been discussed qualitatively in PDM fitting; the explicit demonstration of the full ridge, its escape direction, and its capture of the unconstrained optimum on real PDM data is, to our knowledge, new.

Table 2
Model ladder: nested apparent-parameter models fit to the joint Cr+Fe layer-thickness data. The M3 row is the interior optimum reached from the standard starting point, and it sits on the degenerate ridge itself: the growth prefactor and dissolution rate diverge together to $\sim 10^3$ nm/h while the field parameter collapses to -4.4×10^{-6} nm^{-1}. It attains the lowest data χ^2 in the ladder and the least defensible physics, predicting 155 nm at ten years against M4's 535 nm; it is listed for completeness and rejected on structure, not on fit. The power law appears twice: evaluated at Veile et al.'s published scan-level coefficients, and refit under the same weighted objective (16) as the mechanistic models. For M3 and M5, n_{free} equals the number of data points and dof = 0. ‡The published-coefficient power law is *evaluated*, not fitted, against these six means (its coefficients were determined elsewhere, on individual scans in unweighted linear space), so it costs no degrees of freedom here and its residual dof is 6. No information criterion is computed for that row.

Model	Extra params	n_{free}	χ^2_{data}	dof	A_{bl} (nm/h)
M1 core	—	4	7.15	2	0.7516
M2	$+C_{\text{bl}}$	5	7.15	1	0.7516
M3 (degenerate)	$+C_{\text{bl}}, +C_x$	6	4.11	0	955.8
M4 (accepted)	$+L_{\text{ol},0}$	5	**5.79**	1	**0.7269**
M5	$+L_{\text{ol},0}, +C_{\text{bl}}$	6	5.79	0	0.7269
Power law (published coeffs)	k, n per layer	0‡	20.10	6‡	—
Power law (weighted refit)	k, n per layer	4	7.84	2	—

The trap is resolved by a single additional parameter, the initial outer-layer thickness (M4): a nonzero starting population of Fe-rich outer-layer crystals captures the rapid initial precipitation Veile et al. observe by 72 h. This five-parameter model reduces the data χ^2 from 7.15 (M1) to 5.79, and does so while returning PBR_{eff} to order unity, where M1 needed 2.43 to compensate for the missing initial precipitate population. The six-parameter M5, which adds a free dissolution rate on top of M4, reaches the same χ^2 with that rate indistinguishable from zero; it is carried forward only as the alternative long-time branch it implies (finite saturation instead of unbounded logarithmic growth, Section 3.6), not as a competing fit. For M3 and M5 the free-parameter count equals the six data points and the residual degrees of freedom are zero; their χ^2 is nonetheless nonzero because the objective (16) carries one prior residual in addition to the six data residuals, so the fit is not free to interpolate. No goodness-of-fit statistic is quoted for those rows.

The comparison with the empirical power law is reported both ways. Evaluated at Veile et al.'s published per-layer coefficients the power law gives $\chi^2 = 20.10$ against the joint layer means, but those coefficients were fit to individual scans in unweighted linear space; refit under the weighted objective (16) it reaches $\chi^2 = 7.84$ with four free parameters. On information criteria the two descriptions are then statistically indistinguishable on this dataset (Akaike criterion 15.8 for both; Bayesian criterion 14.8 mechanistic versus 15.0 power law; the small-sample-corrected criterion is undefined for M4 at $n = 6$, $n - k - 1 = 0$, and is 55.8 for the power law). These criteria are quoted as indicative only at this sample size. The mechanistic model's advantage on the calibration window is not fit quality (no comparison on three exposure times could demonstrate that), but the physical coupling it enforces: the constant-volume constraint ties the two layers through a single growth-and-precipitation picture, where the power law treats each layer as an independent curve (Fig. 2); it is this coupling that drives the divergent extrapolations of Section 3.6.

On absolute goodness of fit, $\chi^2 = 5.79$ corresponds to $p = 0.016$ at one residual degree of freedom and $p \approx 0.12$ at three (the effective-dof bracket of Section 2.3). If the mild misfit is real, the reported standard errors of the layer means understate the scan-to-scan variability (consistent with the physical heterogeneity Veile et al. report), and an error-scale correction of $\hat{s} = \sqrt{\chi^2/\text{dof}} = 1.4$–$2.4$ would widen every uncertainty interval below by that factor. Unscaled intervals are quoted throughout; no identifiability verdict changes under the maximal scaling (the sharply identifiable parameters remain so at 2.4× wider intervals, and the unidentifiable ones cannot get worse). The factor is carried forward for one purpose: it sets how far the ten-year band of Section 3.6 can be inflated before the comparison that section rests on would fail, and that comparison survives the inflation.

The recovered kinetics place the apparent growth prefactor at $A_{\text{bl}} = 0.727$ nm/h and the field exponent at $b_3 = -0.0125$ nm^{-1} (-1.25×10^5 cm^{-1}). The recovered $\text{PBR}_{\text{eff}} = 1.096$ carries no prior and is therefore a statement of the likelihood alone. It sits near unity, and roughly a factor of two below the value a chromium mass balance on this alloy implies for an $FeCr_2O_4$ barrier. That comparison does not survive scrutiny in either direction, since the mass-balance

Table 3
Residual and χ^2 leverage decomposition of the accepted M4 fit. σ is the standard error of the layer mean (Table 1). The chromium layer carries 95% of χ^2 and the 168 h chromium point alone carries 80%.

Layer	t (h)	σ (nm)	Model (nm)	Residual (σ)	% of χ^2
Cr (barrier)	72	4.45	41.4	+0.92	14.7
	168	10.82	74.9	−2.15	80.1
	480	3.14	134.8	+0.15	0.4
Fe (outer)	72	64.37	158.9	+0.24	1.0
	168	71.35	195.6	−0.42	3.0
	480	99.59	261.2	+0.21	0.8
Total $\chi^2 = 5.79$				Cr 5.51	95.2
				Fe 0.28	4.8

figure is phase-dependent, and the fitted parameter is not the same quantity once outer-layer loss is admitted; it is taken up in Section 4.

Where the fit statistic comes from

A five-parameter model fitted to six means invites the question of which data points actually carry the fit. Decomposing χ^2 at the M4 optimum answers it (Fig. 2e,f; Table 3). The three chromium points contribute 95% of $\chi^2 = 5.79$ and the three iron points only 5%: the outer layer's scan-to-scan scatter is so large (standard errors of the mean of 32–45% of the mean, against 2–12% for chromium) that it constrains the fit only weakly. Within the chromium set the misfit is concentrated in a single datum: the 168 h point, at -2.15σ, supplies 80% of the total. The residual signs run +, −, + across the three exposures, a systematic pattern rather than scatter, indicating that the *curvature* of the fitted logarithmic law does not track the three measured chromium thicknesses.

Two consequences follow, and both are carried through the rest of the paper. First, the curvature parameter b_3 is what governs the long-time extrapolation of Section 3.6, so a systematic curvature residual on the only three points that constrain it is the dominant structural caveat on the ten-year number, and in our view larger than the parametric uncertainty the bootstrap propagates. Second, the constant-volume coupling and the initial-precipitate parameter $L_{\mathrm{ol},0}$, although motivated by the iron layer, are in practice constrained by it only weakly; the mechanistic reading of the iron exponent in Section 4 should be weighed accordingly. Neither observation is grounds for preferring the power law, whose own free parameters are fitted against the same six means with the same leverage imbalance, but both bound how much any description calibrated on this dataset can claim.

3.2. Parameter identifiability and uncertainty quantification

If three chromium points carry the fit, the question that decides what the fit is worth is which of the five parameters those points actually determine, and the answer governs every extrapolation in Section 3.6, since a parameter the data do not constrain cannot be relied on outside the window that failed to constrain it. Profile-likelihood scans referenced to the joint global minimum (Fig. 3, Table 4) show that the three exposure times sharply constrain three of the five M4 parameters: the growth prefactor, field exponent, and initial outer-layer thickness (maximum $\Delta\chi^2$ of 50, 130, and 58 across their scans). The Pilling–Bedworth ratio is only weakly identifiable (maximum $\Delta\chi^2$ 19.5, interval spanning an order of magnitude), and the initial barrier-layer thickness is formally unidentifiable from the exposure-time data (maximum $\Delta\chi^2$ of 1.0; its quoted interval simply reflects the ± 1.5 natural-log scan range: the parameter is unbounded within the scanned range and is set by the air-film prior). A shallow secondary structure in the growth-prefactor profile at low parameter values is a local optimum of the compensating parameters; it sits more than six $\Delta\chi^2$ units above the global minimum and does not affect the quoted interval. Separately, an explicit search for a dissolution-rate upper bound from the thickness data finds no bound below 1 nm/h: the model's two qualitatively different long-time behaviors remain numerically indistinguishable within the exposure window, a point taken up in Section 3.6.

For a parameter carrying a prior, the profile alone cannot separate data leverage from prior leverage, because both are centered where the prior is. The direct test is to move the prior and see whether the estimate follows (Table 5). L_0 follows it completely: widening its prior fivefold moves the estimate from 1.92 to 1.14 nm (41%), and removing it drives L_0 to zero, the likelihood having no interior optimum at all. $\mathrm{PBR_{eff}}$ behaves in the opposite way. It carries no prior

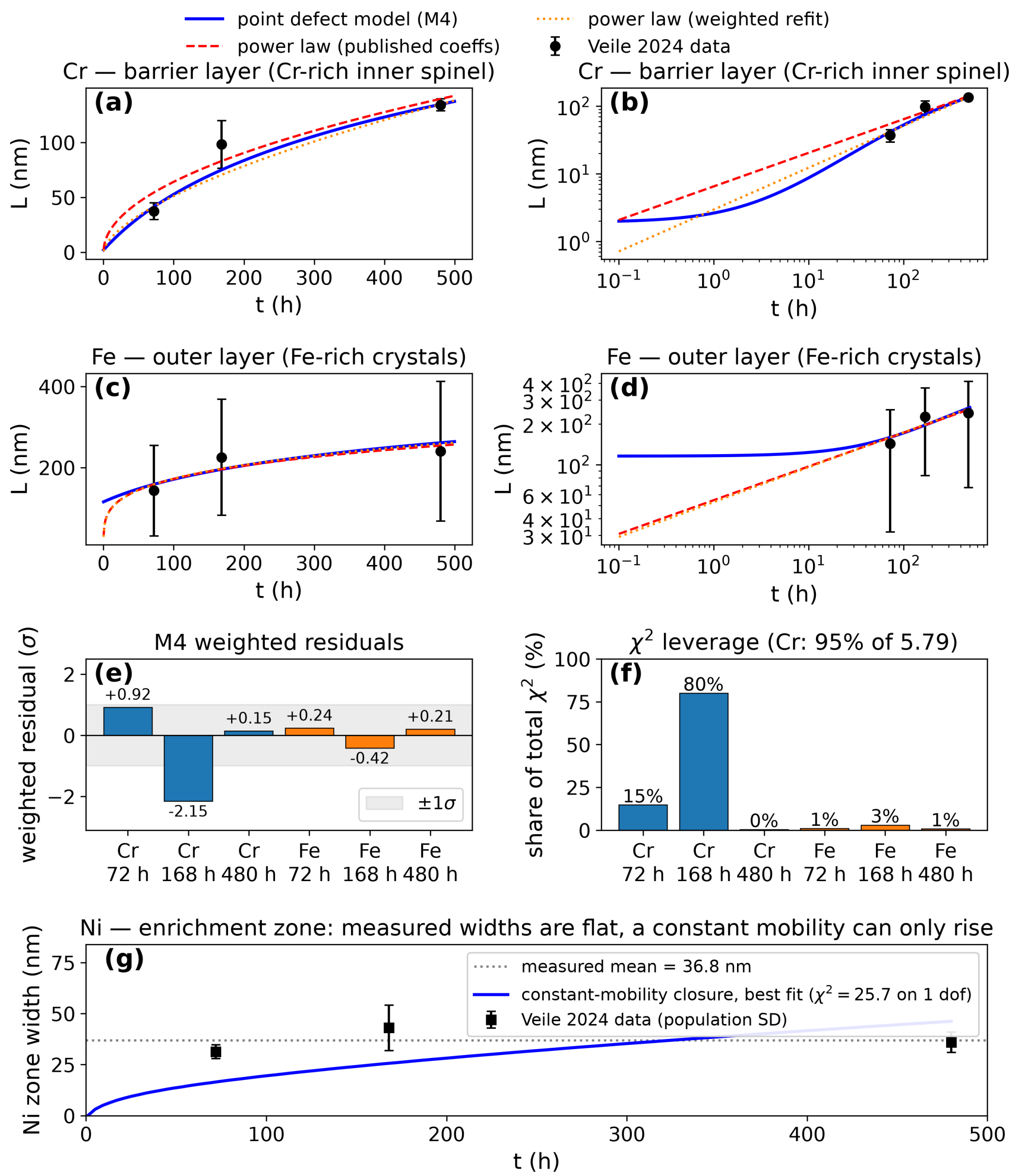

Figure 2: M4 fit and the empirical power law against the measured layer-thickness data. (a, b) Cr barrier layer, linear and log–log axes; (c, d) Fe outer layer, linear and log–log axes. Points with error bars in (a)–(d) are the layer means ± population standard deviations of Table 1; the fits weight residuals by the standard errors $\mathrm{SD}/\sqrt{n}$, so the plotted bars are wider than the quantity actually minimized. Three curves are drawn in each of (a)–(d): the blue solid curve is the M4 mechanistic fit, the red dashed curve the power law evaluated at Veile et al.'s published coefficients, and the orange dotted curve the power law refit under the weighted objective (16). (e) Weighted residuals (model minus measurement, in units of the standard error) of the M4 fit at the six data points, with the shaded band marking $\pm 1\sigma$, and (f) each point's share of the total χ^2; bars are colored by layer in both, blue for chromium and orange for iron. The chromium layer carries 95% of the fit statistic and the 168 h chromium point alone 80%, with the +, −, + chromium residual pattern indicating a systematic curvature miss rather than scatter (Table 3). (g) Ni enrichment-zone width versus exposure time: black squares are the measured widths, the dotted gray line their mean, and the blue curve the best-fit constant-mobility transport closure of Section 3.5 evaluated densely in time. The rejection there is one of shape: the closure can only rise, while the three measured widths are flat within scatter across a factor of 6.7 in exposure time. As in Veile et al.'s own analysis, this zone is not a growing PDM layer, and the fit is shown to display its failure, not as an accepted description; error bars here are population standard deviations, the weighting that fit uses.

Table 4
M4 parameters: fitted value, profile-likelihood 1σ interval, 1000-member bootstrap mean $\pm$ standard deviation, and identifiability verdict. The derived field strength is $\varepsilon_f = 1.7 \times 10^4$ V/cm at the reference transfer coefficient $\alpha_3 = 0.12$; its dominant (systematic) uncertainty is the assumed α_3, discussed in Section 4. †For L_0 the bracket gives the scan-range endpoints, not $\Delta\chi^2 = 1$ crossings: the profile never exceeds $\Delta\chi^2 = 1$ anywhere in the scanned range. No parameter carries a prior except L_0; $\mathrm{PBR_{eff}}$ is determined by the likelihood alone.

Parameter	Unit	M4 fit	Profile 1σ	Bootstrap	Verdict
A_{bl}	nm/h	0.7269	[0.626, 0.845]	0.735 ± 0.122	identifiable
b_3	nm^{-1}	−0.01249	[−0.0145, −0.0108]	-0.01243 ± 0.00202	identifiable
$\mathrm{PBR_{eff}}$	—	1.096	[0.245, 2.32]	1.169 ± 0.931	weakly identifiable
L_0	nm	1.9237	[0.429, 8.20]†	1.923 ± 0.0358	not identifiable (prior-set)
$L_{\mathrm{ol},0}$	nm	115.58	[25.79, 210.6]	110.5 ± 74.7	identifiable

Table 5
Prior-leverage test. Each row refits M4 with the Gaussian log-space prior widths of Eq. (16) modified as shown ("off" = prior removed). A data-determined parameter does not move when its prior is relaxed; a prior-set one drifts or loses its optimum.

Prior setting	$\sigma_{\ln \mathrm{PBR}}$	$\sigma_{\ln L_0}$	$\mathrm{PBR_{eff}}$	L_0 (nm)	χ^2_{data}
As fitted (baseline)	—	0.60	1.096	1.924	5.79
Impose PBR 1.05, $\sigma = 0.20$	0.20	0.60	1.051	1.924	5.79
Impose PBR 1.05, $\sigma = 1.00$	1.00	0.60	1.069	1.924	5.79
L_0 prior ×5	—	3.00	1.097	1.144	5.71
L_0 prior off	—	—	1.098	0.000	5.58
Both (impose PBR, L_0 off)	0.20	—	1.051	0.000	5.59

here, so the test is run in reverse: imposing the prior this analysis previously used, centered on 1.05 with $\sigma_{\ln} = 0.20$, moves the estimate only from 1.096 to 1.051 (4.1%) and leaves the data χ^2 at 5.79. The likelihood therefore does define an interior optimum for this parameter, and the former prior was displacing it by about 4% rather than creating it. This is the test that licenses reading $\mathrm{PBR_{eff}} \approx 1.10$ as a weak but genuine data statement while reading L_0 as prior-set.

The 1000-member parametric bootstrap cross-validates these conclusions by an independent method (SI, Fig. S5), and the two weakly constrained parameters fail it in opposite directions. For the sharply identifiable growth prefactor and field exponent the bootstrap and profile intervals agree closely. For L_0 the bootstrap spread (± 0.036 nm) is far narrower than the profile interval; this is not a tighter constraint but an artifact of the bootstrap perturbing only the data while the profile explores the full parameter range: when the data have no leverage, every resampled refit collapses onto the same prior. $\mathrm{PBR_{eff}}$, now unconstrained by any prior, shows the opposite signature: a bootstrap standard deviation of 0.93 on a central value of 1.10, with the estimate collapsing toward zero in a sizeable minority of replicates (16th percentile 5×10^{-9}). Its profile interval [0.245, 2.32] and its bootstrap spread agree that three exposure times pin this parameter only to within about an order of magnitude. The initial outer-layer thickness sits between these cases: its profile is sharp ($\Delta\chi^2 = 58$) but its bootstrap is not (110 ± 75 nm, 16th percentile 17 nm), because $L_{\mathrm{ol},0}$ and $\mathrm{PBR_{eff}}$ trade off against each other in the outer-layer data and a resampled replicate can move along that trade-off at little cost in χ^2. The profile-likelihood interval, not the bootstrap spread, is the honest uncertainty statement for L_0; for $\mathrm{PBR_{eff}}$ the two methods agree that the constraint is weak; and for $L_{\mathrm{ol},0}$ the wider bootstrap should be preferred, since it is the one that samples the trade-off. All three verdicts are consequences of the same shortage of exposure times.

3.3. Validating the neural inversion, and a failure mode specific to sparse data

Every verdict above was read off a single converged optimum, which raises the question of how far that optimum depends on the machinery that produced it. Because the reduced kinetics have a closed-form solution, this problem admits something that most physics-informed inverse problems do not: an exact reference against which the neural inversion can be graded rather than merely trusted. This section uses that reference twice: once to establish that the hard-mode inversion is interchangeable with a classical fitter, and once to expose a soft-mode failure that the training loss does not reveal.

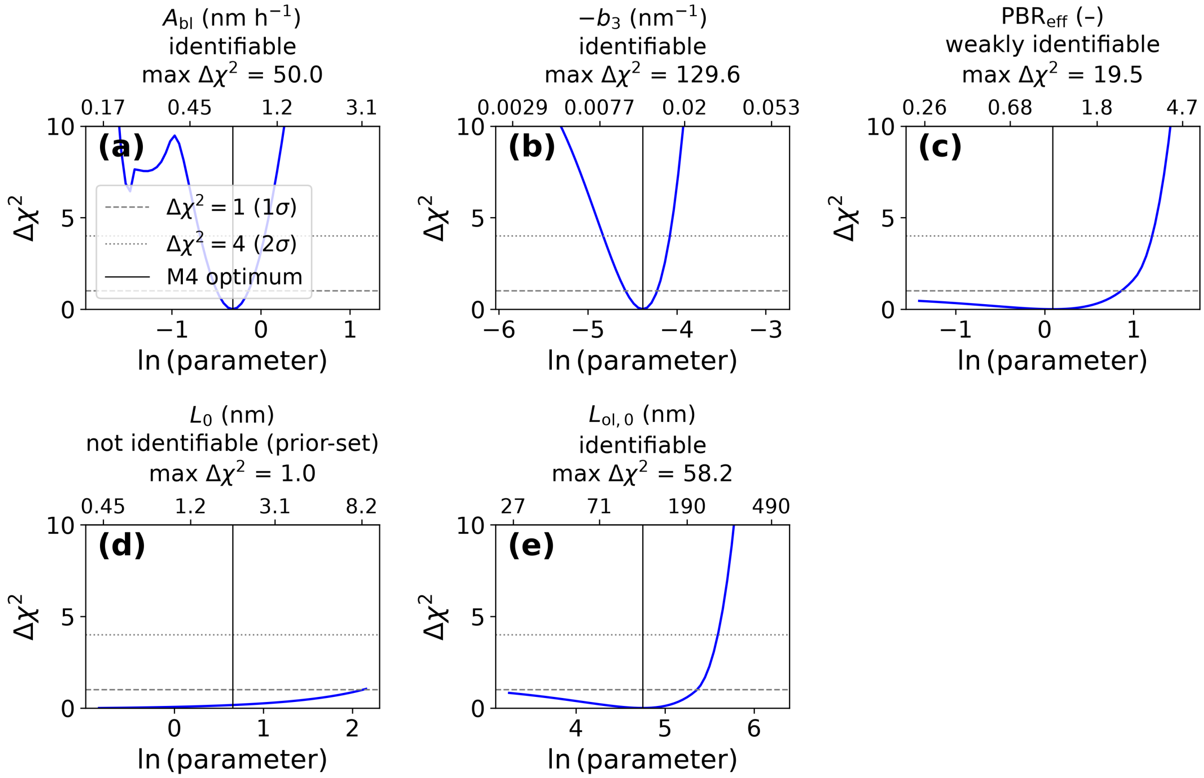

Figure 3: Profile-likelihood identifiability map: $\Delta\chi^2$ over the data residuals versus each M4 parameter, with the 1σ ($\Delta\chi^2 = 1$) and 2σ ($\Delta\chi^2 = 4$) thresholds marked. Panels are the five M4 parameters in the order of Table 4: (a) A_{bl}, (b) $-b_3$ (the magnitude is plotted, b_3 itself being negative throughout), (c) $\mathrm{PBR}_{\mathrm{eff}}$, (d) L_0, (e) $L_{\mathrm{ol,0}}$. In every panel the blue curve is the profile, obtained by fixing the named parameter at each grid value and re-optimizing all the others; the vertical solid line marks the M4 optimum, and the dashed and dotted horizontal lines the two thresholds. The lower abscissa is the fitted coordinate, the natural log of the parameter magnitude; the upper abscissa is the parameter itself, in the units given in the panel title. Each panel title also carries the resulting verdict and the maximum $\Delta\chi^2$ attained over the scanned range: a profile that stays flat is the signature of a parameter the data do not determine. The secondary structure at low A_{bl} in (a) is re-optimization of the remaining parameters along a shallow direction; it lies far above both thresholds and does not affect the interval.

Applied to the real dataset, the hard-mode neural inversion recovers the deterministic M4 optimum to within 0.6% on every parameter, and its trajectory χ^2 of 5.79 matches the classical baseline exactly (SI, Table S9). Its recovery behavior was first validated on synthetic data generated from known ground-truth kinetics with 2% relative noise at the same exposure times: recovery deviations are 1.37% (growth prefactor), 3.39% (field exponent), 0.59% (Pilling–Bedworth ratio), 11.06% (initial barrier-layer thickness), and 0.16% (initial outer-layer thickness). The one large deviation lands on exactly the parameter the profile-likelihood analysis flags as unidentifiable, a direct cross-validation of the identifiability finding by an independent method. A 50-member subset of the bootstrap re-solved through the hard-mode inversion gives spreads statistically consistent with the 1000-member deterministic ensemble (growth prefactor 0.74 ± 0.12 deterministic, 0.76 ± 0.14 physics-informed), confirming the two fitters are interchangeable. Both neural backbones also pass the 0.5% forward acceptance criterion under identical settings (relative L^∞ errors 0.14% and 0.12% for the two thickness fields with the multilayer-perceptron backbone, 0.11% and 0.04% with the Kolmogorov–Arnold-network backbone), a parity result: on this problem, backbone choice does not materially change forward accuracy.

The soft-mode inversion behaves differently. Trained on the same data, the joint field-and-parameter optimization settles into an equilibrium at which the data loss is essentially zero while the governing equation along the recovered trajectory is not satisfied to the precision the low data loss suggests. Re-solving the recovered parameters through the exact forward solution exposes the inconsistency directly: the trajectory's own χ^2 is 0.21, but the same parameters evaluated through the exact solution give $\chi^2 = 16.6$ (SI, Fig. S6 and Table S9), a factor of roughly 80 apart. Increasing the fixed weight on the physics residual does not resolve this, because the adaptive loss aggregator renormalizes relative term weights during training; an independent reweighting toward the data term reproduces the same signature (trajectory 0.67 versus closed-form 15.6). The specific values vary from training run to run, but the one-to-two-orders-of-magnitude signature is stable (SI, Section S4.4). We term this failure mode F-1 and attribute it to the gradient geometry near the data manifold: once the data loss is near zero, the physics-residual gradient is too small to pull the trajectory off that manifold. This mechanism, and the expectation that dense data would prevent the imbalance, are offered as a working explanation consistent with the observations rather than as a demonstrated result.

The practical fix is the hard-mode construction, available for the zero-dimensional kinetics but not, in general, for the spatial extension. For soft-mode inversions we therefore apply, and propose for reporting alongside any physics-informed inverse result, a self-consistency check: re-solve the recovered parameters through an independent physics-exact forward solve (closed form, Runge–Kutta, or Newton iteration) and report the resulting χ^2 next to the trajectory χ^2. In our experience this is the only test among those we tried that reliably distinguishes a trustworthy soft-mode fit from an F-1 failure without requiring a hard-mode alternative to exist; its acceptance rule is a stated heuristic (SI, Section S4.4).

3.4. Spatial extension: defect transport and composition

Veile et al.'s line scans resolve full composition-versus-depth profiles that the zero-dimensional fit never uses. We therefore extend the model spatially (resolving oxygen- and cation-vacancy transport across the barrier layer under the already-identified kinetics), and use the resulting composition predictions as an independent cross-check against data the fit never saw. The transport problem admits no differentiable-integrator shortcut of the kind the zero-dimensional kinetics allow, so the spatial solves use the soft-mode field representation; it is verified against an independent Newton solve on the same node set, which for this steady problem is both cheaper and more accurate, so the comparison below grades the neural solver rather than relying on it. Every kinetic parameter entering this section is frozen at its zero-dimensional value.

That cross-check is only worth running on quantities the measurement could resolve, so before any spatial inversion a synthetic identifiability gate asks which spatial quantities the dataset could constrain at all, using an instrument model with position jitter, compositional noise, and, critically, the scan-to-scan physical heterogeneity Veile et al.'s replicate scans exhibit (SI, Section S2.5; omitting the heterogeneity term produces a spuriously optimistic verdict for every parameter). Twenty replicate synthetic experiments (SI, Table S10) show that the barrier-layer thickness, interfacial transition widths, nickel-zone amplitude and width, and in-layer chromium fraction are identifiable (replicate relative standard deviations 2–19%), while the in-layer defect-transport composition gradient, the one quantity the spatial extension uniquely adds, is weak (47%, with a minimum statistically detectable gradient of 0.93 absolute weight percent across the layer). The spatial model's inverse scope is restricted accordingly, and the defect-transport gradient is treated as a forward prediction, not a fitting target.

The steady-state transport problem, solved by Newton iteration on a spectral collocation grid and by an NSEM forward solve, agrees with its analytic reference to 8.9×10^{-16} (oxygen vacancies) and 2.2×10^{-13} (cation vacancies); the NSEM solve agrees with the Newton reference to 2.5×10^{-4} and 5.4×10^{-5} relative error (SI, Fig. S7). The verification also establishes a structural identifiability result: the steady nondimensional profile shape depends only on the Péclet-type group $\mathrm{Pe} = |z|\gamma\varepsilon_f L$ (in which z is the charge number of the migrating defect, L the barrier-layer thickness, and γ and ε_f are as above), which the zero-dimensional fit already fixes, and not on the defect diffusivity. This was confirmed numerically, not only by inspection of the equations: solving the identical problem with the diffusivity varied over four orders of magnitude leaves the profile shape unchanged to machine precision, while the absolute concentration scale, invisible to a composition measurement such as EDX, scales inversely with diffusivity exactly as the equations predict. Quasi-steady composition measurements therefore cannot constrain the defect diffusivity; only genuinely transient early-time composition data would carry diffusivity sensitivity, through the migration transit times quantified below. This structural argument independently corroborates the statistical gate above.

Everything above treats the defect profile as quasi-steady, which is an approximation every zero-dimensional PDM application makes implicitly and none, as far as we are aware, has quantified: the profile is assumed to equilibrate instantaneously to its steady shape as the layer grows. The transient extension, solved on a moving Landau-transformed domain tracking the growing layer, measures the error that assumption incurs. At the end of the 480-h window the fast oxygen-vacancy species (migration transit time of order 5 h) lags its quasi-steady shape by 0.69%, and the slower cation-vacancy species (transit time of order 36 h) by 5.79% (SI, Fig. S2), both errors scaling with the ratio of transit time to growth timescale as the physics predicts. Both are inside the acceptance tolerances used here, so the quasi-steady treatment of the composition prediction below is licensed rather than assumed.

The composition-map prediction uses no free parameters, and its two components test different things. The predicted chromium plateau inside the barrier layer, 46.7 weight percent against a measured plateau of approximately 45 weight percent, follows from spinel stoichiometry alone, using the $FeCr_2O_4$ assignment (Ziemniak and Castelli, 2003) Veile et al. report as one of several candidate chromium-rich spinels (alongside Cr_2O_3, Fe_2CrO_4, and $NiCr_2O_4$) consistent with their X-ray photoelectron spectroscopy results, and therefore validates a plausible phase assignment, not the kinetics. The kinetic content is in the layer widths, which the frozen kinetics match to within +22%, −20%, and +3% at the three exposure times (Fig. 4), agreement at the tens-of-percent level from a zero-free-parameter prediction, not a quantitative reproduction, with widths extracted from prediction and measurement by the same criterion Veile et al. use (chromium trace exceeding its far-field baseline by three weight percent). The predicted in-layer defect gradient (1.45, 0.96, and 0.46 absolute weight percent at 72, 168, and 480 h) straddles the 0.93 weight percent detection floor from the identifiability gate: marginally detectable at the earliest exposure, undetectable by the latest.

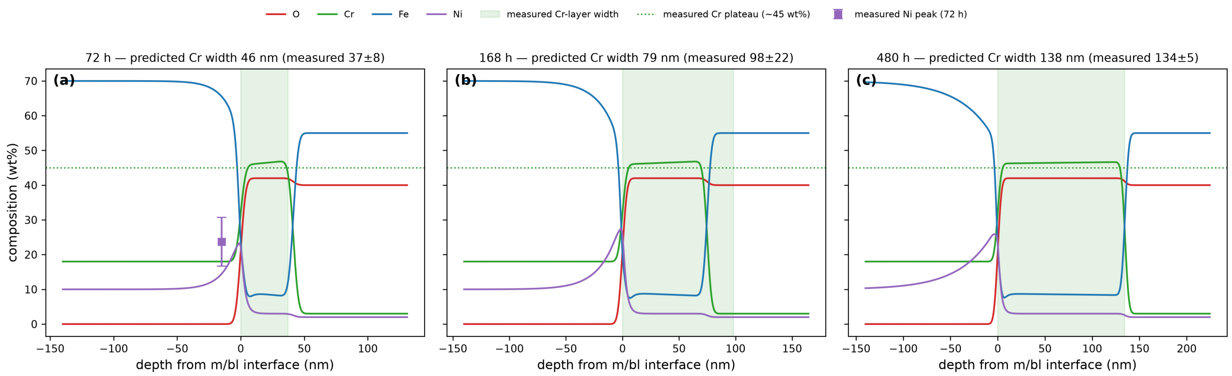

Figure 4: Zero-free-parameter composition-map prediction against Veile et al.'s measured EDX features at (a) 72, (b) 168, and (c) 480 h. Curves are predicted composition profiles (O, Cr, Fe, Ni; weight percent) versus depth from the metal/barrier-layer interface; overlays mark the measured features compared: Cr-layer width (shaded band), Cr plateau level (∼45 wt%, dotted), and the 72-h Ni-peak amplitude (23.7 ± 7.0 wt%, error bar). Full digitized EDX traces are not available, so the comparison is feature-based.

3.5. The nickel enrichment zone

The nickel zone is the one feature of the duplex scale that the identified kinetics do not describe, and the reason is instructive. An exact moving-frame nickel-transport model with a constant effective mobility (SI, Section S3), fitted to the measured zone widths and the 72-h peak amplitude, is *rejected* by the data: $\chi^2 = 25.7$ on one degree of freedom. The rejection is one of shape, not scale: the model predicts a zone width growing monotonically with exposure, while the measured widths are approximately flat (31, 43, 36 nm at 72, 168 and 480 h; Fig. 2g). Because parameter values and nested-model $\Delta\chi^2$ comparisons obtained under a structurally rejected model are not interpretable as estimates or as evidence, the fitted values and the nested-model comparison are reported in the SI (Section S3.1, Table S6) and are not carried into the Discussion as findings.

Two statements about the nickel zone do survive independently of that fit, and they are the section's result. First, the measured widths are flat within scatter across a factor of 6.7 in exposure time, which no constant-mobility transport model of this class can produce: any such model predicts a width growing as the square root of time or faster. Second, any effective mobility consistent with the observed zone scale (a zone tens of nanometers wide developing over hundreds of hours) is of order 10^{-17} cm^2/s, which exceeds extrapolated lattice self-diffusion of nickel in austenite at 513 K by many orders of magnitude. This is a scale argument, independent of the fitted value: the zone-forming transport must be short-circuit or defect-flux mediated rather than lattice-diffusive. Taken together, the two point to a time-decaying or interface-coupled mobility.

That mechanism is exactly what three exposure times cannot resolve. Two physically motivated one-parameter extensions were tested, a defect-flux-coupled effective mobility tied to the identified growth kinetics and a finite-capacity interfacial trapping closure, and both fail a twenty-replicate synthetic identifiability gate before touching real data (SI, Table S7): the flux-coupled closure's new parameter is recoverable but destabilizes the already-identified transport parameters past threshold, and the finite-capacity closure's parameter is not recoverable at all (SI, Section S3.2). A forward-only scan across the flux-coupled closure's plausible range confirms no single value is simultaneously shape-correct and a quantitative improvement. Resolving the nickel zone is therefore left as a target for denser exposure-time data, and is one more instance of the exposure-time limitation that recurs throughout this paper.

3.6. Long-time extrapolation and operating-envelope sensitivity

The identified kinetics are wanted for a horizon no exposure in this dataset reaches, and it is there that the mechanistic and empirical descriptions part company. The result of this section is that separation rather than the thickness it brackets: the point estimate inherits every identifiability limitation cataloged above, while the separation survives them.

At ten years (a factor of roughly 180 beyond the 480-h calibration window), the identified model predicts a barrier-layer thickness of 535 nm, with bootstrap 68% and 95% intervals of [482, 606] and [447, 704] nm (Fig. 5). These bands quantify parametric uncertainty conditional on the M4 model structure and on the reported standard errors; under the maximal error-scale correction of Section 3.1 ($s = 2.4$, applied in log space) the 95% interval widens to approximately [340, 1040] nm. Extrapolating Veile et al.'s published power-law coefficients over the same horizon instead predicts 1853 nm (a factor of 3.5), and the weighted-refit power law 3375 nm (a factor of 6.3; bootstrap 68% interval on the ratio [5.6, 7.0]). Either power-law prediction lies outside the mechanistic band even after the maximal inflation. The divergence is the practical consequence of the constant-volume-coupled growth law saturating logarithmically while the power law, fit independently to each layer, extrapolates indefinitely.

The model's own two long-time branches, unbounded logarithmic growth (M4) versus eventual saturation (M5), track within about five percent of each other out to 10^5 years, which is well inside the 2–12% standard error of the chromium layer means: no exposure experiment at the precision this technique delivers could distinguish them from thickness data, the prediction-space restatement of the dissolution-rate unidentifiability of Section 3.2. Discriminating mechanistic from power-law extrapolation is, by contrast, feasible. Under a conservative scatter model for a future campaign (the worst observed chromium scan-level relative standard deviation, 22%, averaged over three scans, giving a 12.7% standard error), the separation between the two predictions first exceeds twice that scatter at 2546 h with the published coefficients (1437 h refit) and three times at 4014 h (2020 h refit). These thresholds compare the separation to future measurement scatter alone, and so understate the required exposure. Adding the mechanistic prediction's own parametric uncertainty (the 1000-member bootstrap propagated to each exposure time) and, for the refit, the power-law fit uncertainty (an equivalent bootstrap of the same weighted objective) in quadrature moves the 2σ threshold to 2988 h with the published coefficients and 1928 h with the refit, and the 3σ threshold to 5129 h and 5493 h respectively. No fit uncertainty is carried on the published-coefficient branch, since those coefficients were determined elsewhere and no covariance for them is reported; that branch is correspondingly optimistic at 3σ and the refit branch is the one to plan against.

The 3σ figures carry a qualification the 2σ figures do not, and it is the same lesson the rest of the paper reaches by other routes. Once the power law's own extrapolation uncertainty is carried, that uncertainty grows in proportion to the prediction it attaches to, so the separation measured in standard deviations is not monotone in exposure time. Against the refit it crosses 3σ at 5493 h, peaks at 3.18σ near 12000 h, and falls back through 3σ at 34400 h to 2.89σ by 50000 h. Three-sigma discrimination against the refit is therefore available in a window of roughly 5500 to 34000 h and never afterwards, no matter how long the exposure runs; the 2σ criterion, by contrast, once met is never lost. A designed exposure of roughly 3000 h is the realistic requirement for 2σ discrimination against the full error budget, and 3σ requires 5000–6000 h and must be placed inside that window rather than simply extended toward it. The scatter-only numbers should be read as the optimistic bound they are.

The separation above is established at the one operating condition the data were collected at, which leaves open how much of it survives a change of condition. Because the dataset was collected at a single temperature and oxygen level, the model's sensitivity is mapped across a scanned operating envelope of 200–285°C at 7 MPa and 0.01–8 ppm dissolved oxygen, deliberately extended beyond normal-water-chemistry oxygen levels (~0.2–0.4 ppm) to bound off-normal chemistry. The upper temperature bound is the liquid-phase stability limit at 7 MPa ($T_{sat} \approx 286$°C),

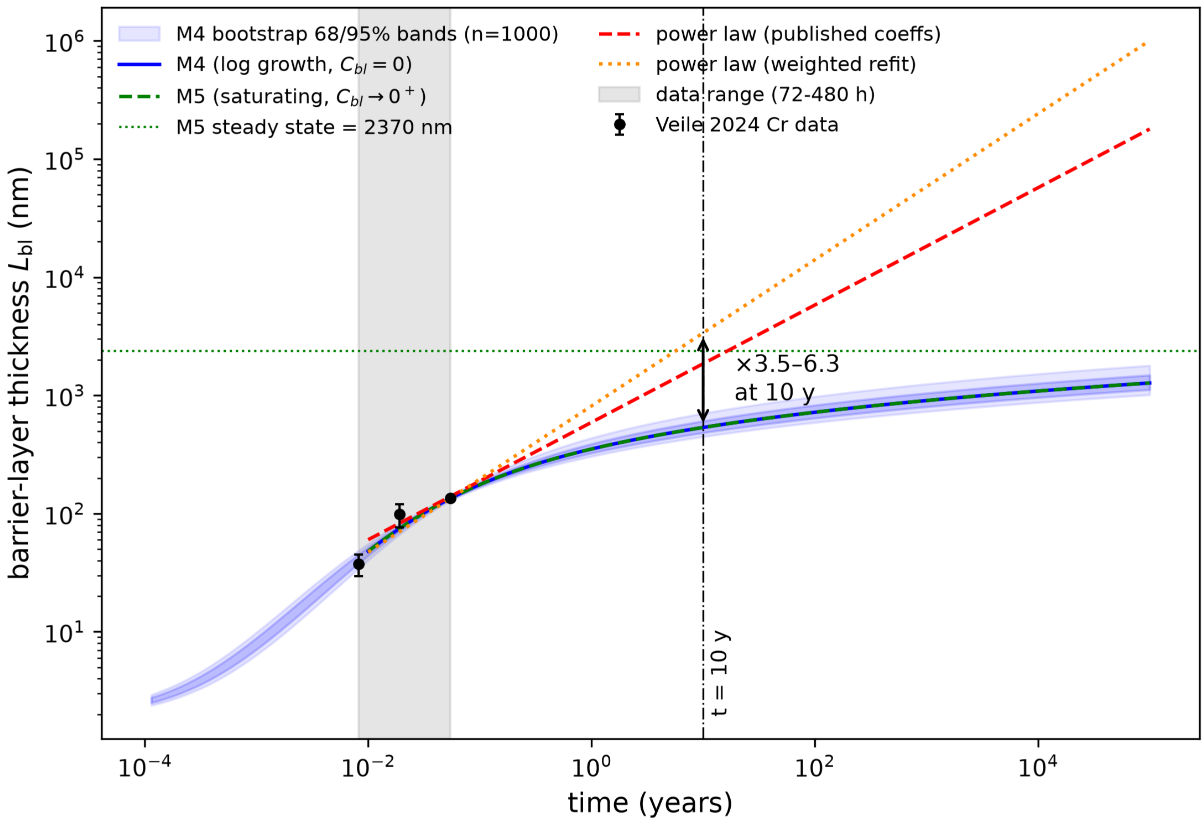

Figure 5: Long-time prediction with propagated parametric uncertainty: M4 (blue solid, logarithmic growth) versus M5 (green dashed, saturating branch, with its steady-state thickness marked by the green dotted horizontal line) versus the two power-law variants (red dashed at the published coefficients, orange dotted for the weighted refit), to 10^5 years on log–log axes. Shaded bands are the 68% and 95% intervals of the 1000-member parametric bootstrap propagated through the M4 closed form, conditional on the M4 model structure and the reported standard errors; the M5 curve is indistinguishable from M4 at all feasible exposures. The three digitized chromium measurements are plotted as black circles and the 72–480 h calibration window is shaded gray, so the interval the extrapolations leave is visible rather than implied. The vertical line marks the ten-year horizon, and the arrow there is the separation the paper's conclusion rests on: the two power-law variants lie a factor of 3.5 to 6.3 above the mechanistic barrier layer and outside its 95% band.

so nominal BWR coolant temperature ($\approx$ 288°C at operating pressure) sits essentially at the edge of the scanned box, not comfortably inside it. The temperature channel follows the rate-constant structure of the supercritical-water formulation (Li et al., 2020): the rate-determining constant scales as $\exp[-\alpha_3 \Delta G_R^0 / R\,(1/T - 1/T_0)]$, where ΔG_R^0 is the standard Gibbs energy of the growth reaction, R the gas constant, T the absolute temperature and $T_0 = 513$ K the calibration temperature, so that the effective Arrhenius activation energy is $\alpha_3 \Delta G_R^0$; with the standard Gibbs energy of the growth reaction unmeasurable from single-temperature data, ΔG_R^0 must be scanned explicitly. The scan is set by the effective activation energy it implies rather than by ΔG_R^0 itself: at $\alpha_3 = 0.12$ the range 25–417 kJ/mol corresponds to $\alpha_3 \Delta G_R^0 = 3$–50 kJ/mol, which spans both the low values implied by transferring a supercritical-water transfer coefficient and the several tens of kJ/mol reported for weight-gain kinetics of austenitic steels in high-temperature water. Scanning only the lower part of that range would build the robustness conclusion on the assumption most favorable to it. A field-attenuation channel is tied to the same temperature dependence, and the dissolved-oxygen channel is an ideal-Nernst shift of the growth prefactor, which is an explicit lower bound on the true corrosion-potential response (Section 4).

Table 6
Operating-envelope spread of the predicted barrier-layer thickness over the scanned box (200–285°C at 7 MPa, 0.01–8 ppm O_2), as a function of the assumed effective Arrhenius activation energy $\alpha_3 \Delta G_R^0$. The first three rows are the originally scanned range; the last two extend it to the values reported for weight-gain kinetics on austenitic steels in high-temperature water.

ΔG_R^0	$\alpha_3 \Delta G_R^0$	L_{bl}(480 h)		L_{bl}(10 y)	
(kJ/mol)	(kJ/mol)	range (nm)	factor	range (nm)	factor
25	3	123–147	1.20	491–583	1.19
50	6	119–151	1.27	487–587	1.21
100	12	112–159	1.42	478–597	1.25
250	30	93–185	1.99	452–627	1.39
417	50	73–214	2.92	423–660	1.56

Across this envelope (Table 6; line sweeps in the SI, Fig. S8), the predicted barrier-layer thickness varies by a factor of 1.20 to 2.92 at 480 h and 1.19 to 1.56 at ten years, depending on the scanned effective activation energy; temperature dominates through the combined Arrhenius and field-attenuation channels, and its sensitivity grows with activation energy, while three decades of dissolved oxygen shift the prediction by under six percent at 480 h and under two percent at ten years, and do so identically at every activation energy, the oxygen channel being a prefactor shift. Two features of this result matter more than the headline range. First, the sensitivity is strongly conditional on the activation energy: over the low 3–12 kJ/mol sub-range the ten-year envelope factor is a nearly negligible 1.19–1.25, but at 50 kJ/mol it is 1.56, and the 480-h factor roughly doubles from 1.42 to 2.92. Second, the ten-year factor is consistently smaller than the 480-h factor, because logarithmic growth compresses prefactor differences at long times: the robustness of the *long-time* prediction is a property of the growth law, and is not inherited from a robustness of the short-time behavior. The long-time prediction is therefore moderately, not strongly, sensitive to operating condition across the full plausible activation-energy range, and a factor of about 1.6 should be carried alongside the parametric band of Fig. 5 rather than the 1.25 the narrow scan alone would suggest.

4. Discussion

The discussion falls into two halves. The first asks what the identified parameters mean: what the empirical exponents become under a mechanistic reading, what the Pilling–Bedworth ratio can be compared against, and what the nickel zone implies about transport at the receding interface. The second bounds how far those readings can be taken: what is and is not attributable to the alloy, what the field strength requires from outside the data, whether the closure that defines it survives a self-consistent solve, and what the calibration environment leaves untested.

The mechanistic model supports a physical reading of Veile et al.'s fitted exponents that the power-law description cannot offer. The near-parabolic chromium exponent ($n \approx 0.50$) is the signature of a barrier layer growing under near-zero dissolution in ultrapure water; the sub-parabolic iron exponent ($n \approx 0.22$) reflects deposition constrained by constant-volume coupling to the barrier layer from a substantial initial precipitate population, rather than independent iron-layer kinetics. This reading is available regardless of fit quality, and it is the coupling behind it, not χ^2, that separates the two descriptions' ten-year predictions by a factor of 3.5 to 6.3.

That coupling is carried by a single parameter, and it is worth asking what its value can be checked against. The identified $\mathrm{PBR}_{\mathrm{eff}} = 1.096$ is one of the two quantities the exposure-time data constrain only weakly, and it is worth being explicit about what it does and does not agree with. An earlier version of this analysis placed a prior on it centered on 1.05 and described that center as the stoichiometric spinel/magnetite value. It is not. The constant-volume closure that yields 1.05 requires the barrier layer to draw approximately 30 at% chromium from an alloy containing 18.8 at%, which the alloy cannot supply. A chromium mass balance on X6CrNiNb18-10 with an $FeCr_2O_4$ barrier gives $\mathrm{PBR}_{\mathrm{eff}} = 2.05$ if every iron atom not retained in the nickel-enriched zone reports to the outer layer, and 2.16 if the nickel routing fraction of Section 3.5 is used in place of complete rejection. That second figure inherits the caveat of Section 3.5: the routing fraction comes from a closure the shape test rejects, so it is admissible here only as a bound on the range, never as an estimate. The first figure is the natural point of comparison, but, as the rest of this section shows, it is neither unique nor a bound.

Table 7
Imposing a stoichiometric $\mathrm{PBR_{eff}}$ once the outer-layer loss term C_x is free. Every row is a refit of that C_x-free family at the ratio shown, the first row included, so the first row is not the accepted model but the same family evaluated at the accepted model's ratio; it returns $C_x \approx 0$, recovering the accepted fit. Every larger ratio is reached at no cost in χ^2, paid for by outer-layer loss and a smaller initial precipitate population. The accepted model has $\chi^2 = 5.79$, $C_x = 0$ and $L_{\mathrm{ol},0} = 115.6$ nm.

$\mathrm{PBR_{eff}}$ imposed	χ^2	$\Delta\chi^2$	C_x (nm/h)	$L_{\mathrm{ol},0}$ (nm)
1.10 (fitted value imposed)	5.784	−0.007	−0.025	120.2
2.05 ($FeCr_2O_4$)	5.671	−0.120	−0.241	93.6
2.16 ($FeCr_2O_4$, Ni routed)	5.661	−0.131	−0.264	90.7
2.43 (M1/M2 optimum)	5.635	−0.157	−0.327	83.0

The likelihood puts the parameter at 1.10, a factor of 1.9 below that figure. That gap is not a conflict, for two independent reasons, and neither of them requires the fit to be wrong.

The first is that 2.05 is not a bound but one value of a phase-dependent quantity. It follows from assigning the barrier the composition $FeCr_2O_4$, and the photoelectron spectroscopy admits three further chromium-rich spinels. Carrying the same mass balance through each of them (using lattice-parameter molar volumes, the nominal alloy composition, and, in one variant, the nickel routing fraction identified in Section 3.4 rather than an assumption of complete nickel rejection) gives production ratios, under complete nickel rejection, of 2.05 ($FeCr_2O_4$), 2.50 ($NiCr_2O_4$), 3.64 (Cr_2O_3) and 0.52 (Fe_2CrO_4); routing nickel at the identified fraction instead moves these to 2.16, 2.50, 3.79 and 0.57. Taken over both routing assumptions the stoichiometrically admissible range is $[0.52, 3.79]$, and the identified 1.10 lies inside it. Two of the eight values lie below 2, so the $FeCr_2O_4$ assignment is one reading of the stoichiometry rather than the reading.

The second reason is that, at any fixed phase assignment, the two numbers are not the same quantity. A mass balance fixes the rate at which barrier growth *produces* outer-layer oxide. The accepted model pins $C_x = 0$, so its $\mathrm{PBR_{eff}}$ measures net *accumulation* instead, and the two differ by whatever oxide leaves for the coolant. Freeing C_x separates them, and doing so shows the data cannot: imposing the $FeCr_2O_4$ value of 2.05 changes χ^2 from 5.79 to 5.67, a *decrease* of 0.12 (Table 7), paid for by an outer-layer loss rate of 0.24 nm/h and a smaller initial precipitate population, 94 nm against the 120 nm the same C_x-free family returns at the fitted ratio (Table 7). Profiling $\mathrm{PBR_{eff}}$ from 0.5 to 8 in this family moves χ^2 by 0.89 in total, so the $\Delta\chi^2 = 1$ interval is the entire scanned range. The reason is structural rather than statistical: with $L_{\mathrm{bl}}(t)$ fixed, the outer-layer solution $L_{\mathrm{ol}}(t) = L_{\mathrm{ol},0} + \mathrm{PBR_{eff}}\,(L_{\mathrm{bl}} - L_0) + C_x t$, with $C_x \le 0$ as in Eq. (11), carries three parameters against exactly three outer-layer observations, so it is saturated and no combination is preferred.

The correct statement is therefore not that the identification disagrees with stoichiometry, but that three exposure times cannot separate oxide production from oxide loss. This also explains an observation that would otherwise look like a loose end: the M1 and M2 variants, which lack the initial precipitate population, reach $\mathrm{PBR_{eff}} = 2.43$; they are simply at a different point on the same flat direction. The resolution makes one falsifiable prediction. If the barrier is $FeCr_2O_4$ and stoichiometry holds, the outer layer must lose about 9 μg of iron per square decimeter per hour, a magnitude consistent with the corrosion-product release reported for stainless steels in high-temperature water (Lister et al., 1987). Nothing measured here demonstrates it: this dataset contains no coolant chemistry, and the outer layer's scan-to-scan scatter is so large that it carries only 5% of the fit statistic (Table 3). A coolant mass balance, or outer-layer thicknesses at more exposure times, would decide it.

The third feature the identified kinetics speak to is the one they do not describe. The order-of-magnitude mobility argument of Section 3.5, which depends on no fitted value and so survives the rejection of the closure that produced it, places whatever transports nickel at the receding metal/barrier-layer interface well above lattice diffusion: it must be short-circuit or defect-flux mediated. That is also the class of mechanism whose mobility would decay as the interface structure coarsens, which is the natural explanation for the flat measured zone widths that defeat the constant-mobility model (Section 3.5). We stop there: distinguishing candidate short-circuit mechanisms requires exposure-time resolution this dataset does not have, and the constant-mobility fit is rejected by the data in shape, so neither its incorporation ratio nor its nested-model comparison against a re-precipitation-only outer-layer supply picture can be offered as evidence either way.

Those three readings are as far as the identified parameters can be taken forward; the rest of this section bounds how far they can be taken at all. One bound concerns the alloy: whether any identified parameter registers the niobium stabilization that is the reason this grade is used in reactor internals. The reaction network fitted here contains nothing niobium-specific, and none of the five identified parameters can be attributed to Nb stabilization on the evidence presented. It is worth being explicit about what that means, since the alloy's stabilization is the reason it is used in reactor internals. The identification is specific to AISI 347 in the sense that the numbers come from measurements on AISI 347, not in the sense that the model structure encodes Nb; every parameter is an apparent quantity for an 18Cr–10Ni austenitic matrix under this environment, and the same structure fitted to 304 or 316 data in the same environment would differ only through the fitted values. Distinguishing a genuine stabilization effect would require the comparison this dataset does not contain: matched exposures of stabilized and unstabilized grades, in which NbC precipitates could be shown to alter local chromium availability at the metal/barrier-layer interface, to nucleate or pin the inner spinel, or to change the initial film that L_0 and $L_{\mathrm{ol},0}$ parametrize. Two of the identified quantities are the natural place such an effect would appear: the growth prefactor A_{bl}, through chromium supply to the growth reaction, and the initial-condition parameters, through surface and precipitate structure. Both are reported here with intervals wide enough that a moderate stabilization effect would be invisible. The claim we make is therefore that this is the first mechanistic identification for this alloy in this environment, not that the mechanism identified is peculiar to the stabilized grade.

A second bound attaches to the field strength. Of the five apparent parameters, the field exponent has the most direct physical reading and demands the most from outside the data to obtain it. Converting the identified field exponent through Eq. (15) requires the growth-reaction transfer coefficient α_3, which single-temperature thickness data cannot supply. At the reference value $\alpha_3 = 0.12$ fit by Li et al. (2020) for a ferritic steel in supercritical water, $\varepsilon_f = 1.7 \times 10^4$ V/cm; over the physically defensible range $\alpha_3 \in [0.1, 0.5]$ the derived field spans 4.1×10^3 to 2.1×10^4 V/cm. The implied potential drop across the 480-h barrier layer, $\varepsilon_f L$, is correspondingly 0.06–0.28 V; mixed-potential estimates of the corrosion potential in oxygenated 288°C BWR water are of order 0 to +150 mV on the standard hydrogen electrode (SHE) scale at 0.2–0.4 ppm O_2 (Macdonald, 1992b; Lin et al., 1996), which favors the lower end of the derived range and is one more reason to treat ε_f here as an order-of-magnitude quantity. Within that reading, the value sits plausibly between the $\sim 10^2$ V/cm of supercritical-water PDM fits at 500°C (Li et al., 2020) and the $\sim 10^6$ V/cm of room-temperature passive films (Cabrera and Mott, 1949); we offer this ordering as a weak plausibility remark only, since the three anchors span different alloy classes and film chemistries, and the nearest-temperature precedents (PDM and mixed-conduction analyses of 304/316 in 280–320°C water (Matthews, Knutsen, Westraadt and Couvant, 2021; Bojinov et al., 2005; Zhao et al., 2024)) do not report directly comparable apparent field strengths.

The identified kinetics sit within the high-temperature-water oxide corpus, tabulated against related systems in the SI (Table S11). The inner-layer thickness trajectory recovered here (41 nm at 72 h to 135 nm at 480 h) is of the same order as inner-layer thicknesses reported for 304/316 stainless steels in oxygenated 288°C water at comparable times (Kuang et al., 2010; Kim, 1999), and the duplex morphology, inner-layer spinel chemistry, and near-parabolic inner growth match the established picture for austenitic steels in this environment (Robertson, 1991; Stellwag, 1998; Ziemniak and Hanson, 2002). The ten-year prediction of 535 nm has no direct validation data for AISI 347; we are not aware of published long-term harvested-component oxide thicknesses for this alloy in BWR conditions, and the prediction should be read as a testable statement, not a validated one.

Those field strengths rest on an assumption the identification itself never tests. Every number derived from b_3 through $b_3 = -\alpha_3 \chi \gamma \varepsilon_f$ inherits the PDM's constant-field closure, and that closure can be tested directly by solving Poisson's equation together with the defect transport instead of imposing the field on it. We have carried out that solve, verifying it by reduction: with the space charge switched off it reproduces the analytic constant-field solution to 1×10^{-15}. Switched on, it does not support the closure (Fig. 6). At the identified kinetics and the transport constants assumed in Section 3.4, the dimensionless space-charge parameter is of order 10^4–10^5 and the corresponding Debye length is approximately 0.06 nm, several times smaller than an interatomic spacing; holding the field within 10% of uniform would require defect diffusivities near 10^{-9}–10^{-10} cm^2/s, some six orders of magnitude above the assumed values, or equivalently defect concentrations below about 10^{14} cm^{-3}. The implication is not that the field distribution is merely non-uniform but that a two-species charged-defect description cannot produce a constant field at the concentrations the model itself implies: charge compensation by electronic carriers, which the PDM does not carry, is required. This leaves the fitted apparent parameters and every prediction drawn from them untouched, since the identification is of b_3 and not of ε_f. What it bounds is the interpretation: ε_f should be read as an apparent field-related grouping, and the numerical field strengths quoted just above carry that caveat.

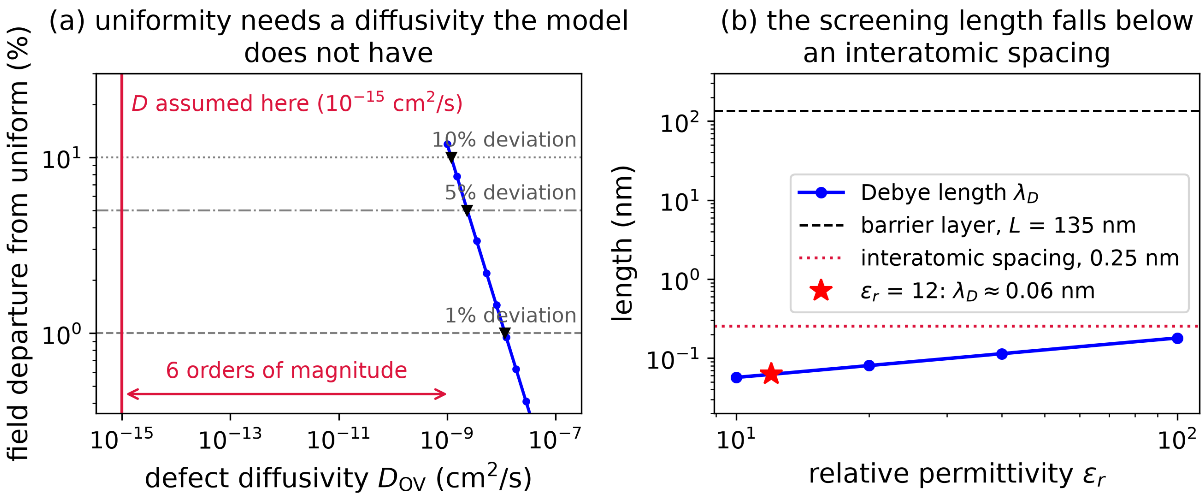

Figure 6: The constant-field closure is not self-consistent at the transport constants the model assumes. (a) Maximum departure of the self-consistently solved field from uniform, against the oxygen-vacancy diffusivity that would be needed to hold it there; markers locate the 1%, 5% and 10% thresholds. The diffusivity actually assumed lies six orders of magnitude below all three. (b) The Debye length against its two yardsticks: the barrier layer it must screen across, and an interatomic spacing. At the permittivity used here it is approximately 0.06 nm, several times smaller than the smaller of the two, which is what makes the closure untenable rather than merely approximate.

A third bound is the calibration dataset itself, which imposes four conditions on everything above, each a property of the measurements rather than of the analysis. First, the kinetics rest on a single temperature and a single dissolved-oxygen level, so the operating-envelope results of Section 3.6 are forward projections under sourced assumptions, not validation against multi-condition data, which does not exist for this alloy. Their temperature channel additionally assumes a single rate-determining step with a temperature-independent field strength; the activation-energy scan bounds the parameter of that channel but not its structure. Their oxygen channel is an ideal-Nernst shift of the growth prefactor, and measured corrosion-potential–oxygen curves in BWR-relevant chemistry are steeper than Nernstian in the parts-per-billion transition region (Lin et al., 1996), so the mapped oxygen sensitivity is a lower bound on the true response. Second, the exposures are unirradiated and static, while radiolysis products, H_2O_2 in particular, are known to alter oxide structure and thickness on stainless steels relative to O_2 at comparable corrosion potential (Kim, 1999), and flow affects outer-layer growth and dissolution through solubility and mass transfer (Lister et al., 1987); neither channel is in the model. Third, surface finish conditions the initial film and the early precipitate population, the parameters L_0 and $L_{\mathrm{ol},0}$ here, and mechanical surface treatments are documented to change oxide properties on 304L in simulated BWR and PWR water (Jaffré, Abe, Ter-Ovanessian, Mary, Normand and Watanabe, 2021), so the identified $L_{\mathrm{ol},0} = 116$ nm is specific to Veile et al.'s coupon preparation. Fourth, EDX-derived layer thicknesses are not total oxide mass, so the composition comparison of Section 3.4 is a shape and plateau-composition cross-check rather than a mass-balance validation, which matters for the iron-release prediction made above, since that prediction is built on thicknesses and would be decided by a coolant mass balance.

A bound of a different kind attaches to the inversion machinery. Nothing in this paper's physical conclusions requires the neural solver: the kinetics have a closed form, the steady transport problem has an analytic reference, and the classical methods used to check the neural results would have produced the same numbers, faster, and in the spatial case to eleven more digits. That is a deliberate property of the test case rather than an oversight, since a validation is possible only where an exact reference exists, but it bounds the claim to this: the framework is trustworthy on real sparse data once the self-consistency check of Section 3.3 is applied, not that it was needed here. The problems that would need it are visible from this study. Two lie inside the present model: solving the potential distribution self-consistently with defect transport instead of imposing the constant-field assumption, and inverting the moving-boundary transient for kinetic parameters and defect fields together, where the classical route requires a remeshed forward solve inside every optimizer iteration. A third is sharper: the nickel zone rejects every constant-mobility description these data can test, so the quantity actually wanted is a mobility *function* of unknown form, which a neural representation can carry as an unknown closure constrained by the transport equation while a classical solver must first commit to a parametric guess.

5. Conclusions

What remains after all four bounds is the paper's recurring finding. Every identifiability limitation encountered here (the zero-dimensional profile analysis, the residual-leverage imbalance, the spatial gate and the nickel-closure gate) traces to too few independent exposure times. The convergence of four unrelated analyses on one cause is the methodological result, and the remedy it implies is concrete: a designed exposure near 3000 h would separate the mechanistic prediction from the power-law extrapolation at 2σ against the full error budget, and would sharpen every weakly constrained parameter at the same time. Neither the cause nor the remedy is specific to this alloy or to the point defect model; both apply to any sparse-data mechanistic identification in which extrapolation beyond the calibration window is the purpose of the fit.

Data availability

The digitized Veile et al. layer-thickness data (Table 1) are transcriptions of published figures, released with the analysis repository below together with provenance notes tracing each value to its source figure; this constitutes reuse of published data requiring citation, not separate data-sharing permission. A committed artifact stands behind every number reported here, and a generated document maps each figure and table to the exact command that produces it, so any reported value can be re-derived individually.

Code availability

All code implementing the deterministic fits, physics-informed forward and inverse solves, identifiability analyses, and spatial extension is released as a standalone, installable Python package (`htw-pdm`) at `https://github.com/Feugmo-Group/x6crninb18-pdm` under the Apache 2.0 license, together with the configuration files and the analysis and figure scripts. The classical fits, the mass balance and the spatial solves depend only on NumPy, SciPy and Matplotlib; the neural inversions additionally require the NSEM solver (Tetsassi Feugmo and Pankaczy, 2026), whose spectral-element-network module (`experimental.models.scen`) is released with PhysicsNeMo, and the repository records the exact revision used here.

Acknowledgements

C.G.T.F. acknowledges the support of the Natural Sciences and Engineering Research Council of Canada (NSERC) [RGPIN-2024-03989]. This research was enabled in part by support provided by SHARCNET and the Digital Research Alliance of Canada.

CRediT authorship contribution statement

Conrard Giresse Tetsassi Feugmo: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing – original draft, Writing – review & editing, Visualization.

Declaration of competing interest

The author declares that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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